MLA0202 – FUNDAMENTALS OF MACHINE LEARNING

COURSE ASSIGNMENT REPORT

Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical-State Reasoning System

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A. Assignment Information

Particular	Details
Department	Artificial Intelligence & Machine Learning
Programme	B.Tech
Course Code & Course Name	MLA0202 – Fundamentals of Machine Learning
Academic Year / Batch	2026
Faculty Name	Dr. Saravanan M.S.
Assignment Title	Design and Development of a Probabilistic Graphical-Model-Based Patient Cohort Segmentation and Clinical-State Reasoning System
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Date of Submission	01/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO3: Analyse unsupervised learning methods, including clustering and dimensionality reduction. CO4: Develop probabilistic graphical models for inference, learning patterns, and analysing sequential data. CO5: Create graphical models to analyse and infer patterns in complex probabilistic systems for structured decision-making.
Bloom's Taxonomy Level	L4 – Analyse, L5 – Evaluate, L6 – Create
SDG Mapping (SDG 1–17)	SDG 3 – Good Health and Well-being; SDG 9 – Industry, Innovation and Infrastructure; SDG 10 – Reduced Inequalities
Industry / Societal Relevance	Directly relevant to hospital clinical-decision-support analytics, patient triage automation, and explainable-AI adoption in healthcare informatics.

B. Assignment Problem / Challenge

A multi-speciality hospital wishes to deploy an analytics module that segments patients into clinically meaningful risk cohorts using their vitals and laboratory profile, and reasons about the probable cause of an abnormal reading together with a patient's evolving clinical state, using probabilistic graphical models. The existing hospital workflow relies on manual review of high-dimensional patient data, which makes cohort identification, dependency reasoning, and clinical-state tracking slow, inconsistent, and difficult to explain to both clinicians and administrators.

This assignment requires the student to analyse the given case and design an unsupervised patient-cohorting and probabilistic-graphical-model-based reasoning pipeline. The pipeline is expected to combine four distinct machine-learning paradigms into a single, coherent, end-to-end clinical decision-support architecture:

- A clustering-based cohort-discovery step that groups patients into a small number of interpretable risk categories from their raw vitals/lab measurements.
- A dimensionality-reduction step that compresses the high-dimensional vitals/lab feature space into a smaller number of well-justified components while retaining clinically meaningful structure.
- A Bayesian network that encodes symptom–risk–disease dependencies and supports exact probabilistic inference over these dependencies given observed evidence.
- A Hidden Markov Model that tracks a patient's evolving clinical state (e.g., Stable → Deteriorating → Critical) from a sequence of noisy vital-sign-derived observations.

The assignment further requires that the outputs of the two graphical models (the Bayesian-network posterior and the HMM state estimate) be combined into a single structured decision rule that recommends one of three clinical actions — escalate, monitor, or discharge — thereby closing the loop from raw sensor data to an explainable, actionable recommendation that keeps a human clinician in the loop at all times, consistent with SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation and Infrastructure) and SDG 10 (Reduced Inequalities), since a well-designed, low-cost, open-source pipeline of this kind can be deployed even in resource-constrained hospital settings and can help reduce disparities in access to timely clinical decision-support.

C. Problem Statement

C.1 Problem / Case / Design Challenge

Clinicians in a busy multi-speciality hospital are required to continuously monitor large volumes of patients, each generating a stream of vital-sign measurements (heart rate, blood pressure, temperature, respiratory rate, oxygen saturation) together with periodic laboratory results (white blood cell count, and other markers) and self-reported or observed symptoms (fever, cough, hypotension, and so on). Under manual review, three distinct cognitive tasks are conflated and performed inconsistently across shifts and clinicians:

- Cohort identification — recognising that a given patient's overall vitals/lab profile resembles a known risk category (e.g., low, moderate, or high risk) so that appropriate ward-level resourcing and monitoring frequency can be assigned.
- Dependency reasoning — inferring, from a partial set of observed symptoms and risk factors, the probability that an underlying condition (e.g., infection, sepsis) is present, in a way that is transparent and defensible rather than a black-box heuristic.
- Clinical-state tracking — recognising, from a noisy time series of vitals, whether a patient's underlying condition is Stable, Deteriorating, or Critical, even when individual readings are ambiguous or contradictory.

Because these three tasks are performed manually and independently, cohort assignment is not reproducible across shifts, dependency reasoning is not explicitly documented (making it hard to audit clinical decisions later), and clinical-state changes are frequently detected late — precisely because a single abnormal reading is easy to dismiss as noise, whereas a probabilistically-tracked trend across several readings is not. The assignment requires the student to replace this manual, ad-hoc process with a formally specified, machine-learning-based pipeline that performs all three tasks in a principled, explainable, and computationally efficient manner, and that feeds its outputs into a single structured decision rule usable by both clinicians and hospital administrators.

C.2 Scope of the Assignment

The scope of this assignment is restricted, per the assignment brief, to the design, implementation, and validation of a synthetic-data pipeline consisting of clustering (K-means and/or Gaussian-Mixture/EM), dimensionality reduction (PCA or Factor Analysis), a Bayesian network (with conditional probability tables and exact inference by variable elimination), and a Hidden Markov Model (with the forward algorithm and/or Viterbi decoding), integrated behind a structured escalate/monitor/discharge decision rule. Deployment concerns such as production-grade software engineering, integration with hospital electronic health record (EHR) systems, and regulatory certification are explicitly out of scope for this course assignment, though they are discussed qualitatively in Section E.7 (Broader Considerations).

D. Requirements and Constraints

The following requirements and constraints, drawn from the assignment brief, govern the design of the Patient Cohort Segmentation and Clinical-State Reasoning System. Only the categories genuinely relevant to this problem are elaborated, consistent with the instruction that only constraints relevant to the particular course/problem need to be considered.

D.1 Functional Requirements

- FR1: Segment patients into a small number ($k = 3$ in this design) of interpretable risk cohorts using clustering on standardized vitals/lab features.
- FR2: Reduce the dimensionality of the vitals/lab feature space while retaining a justified proportion (≥ 90%) of the total variance/structure.
- FR3: Represent symptom–risk–outcome dependencies using a Bayesian network (a directed acyclic graph with conditional probability tables) and compute posterior probabilities of a target condition given arbitrary observed evidence.
- FR4: Track a patient's underlying clinical state over time from a sequence of noisy vital-sign-derived observations using a discrete-state Hidden Markov Model.
- FR5: Translate the Bayesian-network posterior and the HMM state estimate into a single, structured decision recommendation — escalate, monitor, or discharge.

D.2 Performance Requirements

- PR1: Clustering and dimensionality reduction must scale efficiently (near-linear time) to the full patient feature set, so that re-clustering as new patients arrive remains tractable.
- PR2: Bayesian-network and HMM inference must be computable within a few seconds per patient/query so that the system can be used at the point of care.
- PR3: The decision framework must achieve an acceptably low false-negative rate on high-risk/Critical cases, even if this increases the false-positive rate on borderline cases.

D.3 Technical Constraints

- TC1: Patient records must be represented as fixed-length feature vectors suitable for clustering and PCA/Factor Analysis (standardized vitals and laboratory values).
- TC2: The clinical-state sequence must be represented as a discrete-state Hidden Markov Model with a finite observation alphabet.
- TC3: Symptom–risk–outcome dependencies must be represented as a Bayesian network — a directed acyclic graph (DAG) with conditional probability tables (CPTs) — rather than an unstructured rule set.
- TC4: Implementation is restricted to open-source tools: Python, NumPy, pandas, scikit-learn, pgmpy, hmmlearn, and Matplotlib/Seaborn.

D.4 Time and Resource Constraints

- The modelling and implementation must be completed within the time allotted for the assignment, using laboratory computers and freely available open-source software.

- The solution uses a synthetically generated, de-identified dataset and must not require specialised or expensive hardware (e.g., no GPU cluster is required for any of the four modules).

D.5 Reliability and Safety

- False negatives on Critical/Deteriorating states must be minimised even at the cost of a higher false-positive rate — a missed escalation is clinically far more costly than an unnecessary one.
- Every automated recommendation must be explainable to a clinician in terms of the underlying posterior probabilities and state estimates, not as an opaque score.
- Decision recommendations must remain consistent with the underlying graphical-model probabilities at all times — the decision rule is a deterministic, auditable function of the model outputs, never a separate, unexplained heuristic.

D.6 User Requirements

- Clinicians must be able to view a patient's cohort membership, inferred posterior probability, and clinical-state estimate together with a brief natural-language explanation.
- Administrators must be able to view aggregated, structured decision recommendations arising from the graphical models for resource-planning purposes.
- A human clinician must remain in the decision loop at all times — the system is decision-support only and never acts autonomously on a patient.

D.7 Security and Privacy

- Patient vitals, laboratory, and identity data must be handled under de-identification and informed-consent assumptions, consistent with data-protection norms for health information.
- Sensitive clinical information should not be unnecessarily exposed outside the clinical decision-support context — outputs are scoped to authorised clinical and administrative users only.

D.8 Ethical Considerations

- The cohorting and decision framework must not systematically disadvantage any demographic group — no demographic attribute (age, gender, ethnicity) is used as a clustering or Bayesian-network feature in this design, precisely to avoid encoding bias.
- Predictions and recommendations must be used to support, not replace, clinical judgement at every stage of the pipeline.
- The system should maintain accuracy, confidentiality, and fairness in patient outcomes throughout its operational lifetime.

D.9 Sustainability and Scalability

- The pipeline must support additional patients, features, and disease conditions without major architectural redesign — each module (clustering, PCA, BN, HMM) is decoupled behind a well-defined interface.
- The solution should minimise unnecessary computation and storage, an important consideration for hospital IT infrastructure that may be shared across many clinical systems.

D.10 Primary Constraints Summary

Taken together, the primary constraints applicable to this machine-learning problem are: cohort interpretability, justified retained variance/components, correct probabilistic inference, sequential-state accuracy, explainability, data privacy/security, fairness, and scalability. These eight constraints are used throughout Section E as the yardstick against which every design decision is justified.

E. Proposed Work

E.1 Problem Understanding and Formulation

E.1.1 Problem Identification

Manual review of high-dimensional patient data suffers from three concrete limitations that this assignment directly addresses. First, cohort discovery performed by eye across six or more correlated vital/lab features is not reproducible: two clinicians reviewing the same patient chart may reach different risk-tier judgements, particularly for borderline patients whose readings sit between clearly-defined categories. Second, dependency reasoning — for example, judging how likely a patient with fever and an elevated white-cell count is to have an underlying infection or sepsis — is rarely made explicit; clinicians reason informally from experience, which is difficult to audit, teach, or improve systematically. Third, clinical-state tracking from a noisy stream of vitals is inherently a sequential-inference problem, yet manual review typically treats each reading in isolation, which means that a genuinely deteriorating trend spread across several borderline readings can be missed, even though a probabilistic model that integrates evidence over time would flag it.

E.1.2 Expected Outcomes

The pipeline developed in this assignment is expected to deliver the following concrete outcomes:

- A small number of interpretable, clinically nameable risk cohorts (Low-Risk, Moderate-Risk, High-Risk) derived from unsupervised clustering of standardized vitals/lab features.
- A reduced-dimensionality representation of the vitals/lab feature space that retains at least 90% of the original variance while remaining interpretable in terms of clinically meaningful axes (e.g., an 'acute inflammatory severity' axis and a 'cardio-respiratory stability' axis).
- A Bayesian network capable of returning an exact posterior probability of a target disease state given any combination of observed symptom/risk evidence, together with a fully worked example of variable elimination.
- A Hidden Markov Model capable of filtering (forward algorithm) and decoding (Viterbi algorithm) a patient's most likely clinical-state sequence from a stream of noisy vital-sign-derived observations.
- A structured, explainable decision rule that maps the above outputs onto one of three actions: escalate, monitor, or discharge.

E.1.3 Available Data

For this assignment a synthetically generated, de-identified dataset of 300 patients is used, each represented by six standardized vitals/lab features — Heart Rate (HR, beats/min), Systolic Blood Pressure (SBP, mmHg), Body Temperature (Temp, °C), White Blood Cell count (WBC, $\times 10^9$/L), Peripheral Oxygen Saturation (SpO2, %), and Respiratory Rate (RR, breaths/min) — together with three binary symptom/risk indicators (Fever, Elevated WBC, Hypotension) used by the Bayesian network, and a per-patient time series of three vital-sign-derived observations (Normal/Abnormal) used by the Hidden Markov Model. A representative sample of the synthetic dataset is provided in Appendix A; the full 300-row dataset was generated programmatically (see Appendix B) from three Gaussian cluster profiles corresponding to the three intended risk cohorts, which also serves as ground truth for validating the unsupervised clustering step in Section E.5.

E.1.4 Assumptions

- Vital-sign observations are noisy but conditionally independent of one another given the patient's true underlying clinical state — the standard HMM emission-independence assumption.
- The conditional probability values used in the Bayesian network's CPTs are clinically plausible expert-elicited estimates for the purposes of this assignment, not values fitted to a real epidemiological study.
- Cohort structure (the three risk tiers) is assumed stable over the observation window used for clustering; genuine longitudinal drift in cohort membership is out of scope.
- Features are approximately continuous and are standardized (zero mean, unit variance) prior to clustering and PCA, consistent with the Euclidean-distance assumptions underlying K-means and the covariance-based formulation of PCA.
- The three observation categories (Normal/Abnormal) used by the HMM are derived from a simple threshold rule on the raw vitals for the purposes of this assignment; a production system would likely use a richer, possibly continuous-valued, emission model.

E.1.5 Constraints Relevant to Formulation

The formulation above already reflects the constraints identified in Section D: interpretability is preserved by keeping k small (k = 3) and by choosing PCA components with clear clinical loadings; correctness of inference is treated as a first-class deliverable, with every Bayesian-network and HMM computation checked by hand in Section E.5; explainability is built into the decision rule itself, which is a transparent threshold function of the two graphical-model outputs rather than a learned black-box classifier; and privacy is respected by using only a synthetic, de-identified dataset with no real patient identifiers throughout the assignment.

E.2 Application of Course Knowledge

This section identifies the principles, models and algorithms from the course that are applied at each stage of the pipeline, together with the essential mathematical formulation of each.

E.2.1 Clustering — K-means and Gaussian Mixture / EM

K-means partitions n standardized feature vectors $x_1,\ldots,x_n \in \mathbb{R}^6$ into k clusters by minimising the within-cluster sum of squared distances:

$J = \Sigma_{i=1}^{n} \|x_i - \mu_{c(i)}\|^2$, *where c(i) is the cluster assigned to point i and* μ_j *is the centroid of cluster j.*

The algorithm alternates an assignment step (assign each point to its nearest centroid) and an update step (recompute each centroid as the mean of its assigned points) until convergence; because J is non-convex, K-means is run from multiple random initialisations (k-means++ seeding) and the lowest-J solution is retained. The Gaussian Mixture Model (GMM) generalises this by modelling the data as a mixture of k multivariate Gaussians, $p(x) = \Sigma_j \pi_j \mathcal{N}(x; \mu_j, \Sigma_j)$, with mixing weights π_j, fitted by the Expectation-Maximisation (EM) algorithm: the E-step computes the posterior responsibility $\gamma_{ij} = P(\text{cluster } j \mid x_i)$ for every point and cluster, and the M-step re-estimates π_j, μ_j, Σ_j from these responsibilities, iterating until the log-likelihood converges. Unlike K-means, GMM/EM yields a soft, probabilistic cluster assignment, which is useful for flagging patients who sit near a cohort boundary.

E.2.2 Dimensionality Reduction — Principal Component Analysis

Given the standardized feature matrix $X \in \mathbb{R}^{n\times 6}$, PCA computes the eigendecomposition of the sample covariance matrix $\Sigma = (1/(n-1))\, X^T X = V\Lambda V^T$, where the columns of V are the eigenvectors (principal-component directions) and the diagonal of Λ contains the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_6$, each representing the variance captured along that direction. The proportion of total variance explained by the first m components is $\Sigma_{i=1}^{m} \lambda_i / \Sigma_{i=1}^{6} \lambda_i$; the number of retained components m is chosen as the smallest m for which this cumulative proportion exceeds a target threshold (90% in this design), balancing fidelity against interpretability. Factor Analysis, an alternative technique discussed comparatively in Section E.6, instead models each observed feature as a linear combination of a smaller number of latent factors plus feature-specific (uncorrelated) noise, which is often a better fit when individual vitals/lab measurements carry substantial independent measurement noise.

E.2.3 Bayesian Networks and Exact Inference

A Bayesian network is a directed acyclic graph (DAG) whose nodes are random variables and whose edges encode direct probabilistic dependence; each node X is annotated with a conditional probability table (CPT) P(X | Parents(X)). The joint distribution over all n variables factorises as $P(X_1,\ldots,X_n) = \Pi_i P(X_i \mid \text{Parents}(X_i))$, which is exponentially more compact than a full joint table whenever the graph is sparse. Given observed evidence E = e, the posterior probability of a query variable Q is P(Q | E=e) = P(Q, E=e) / P(E=e), computed exactly in this assignment by variable elimination — repeatedly summing (marginalising) out non-query, non-evidence variables from the factorised joint in an order chosen to minimise intermediate factor size — rather than by brute-force enumeration of the full joint distribution.

E.2.4 Hidden Markov Models — Forward Algorithm and Viterbi Decoding

A Hidden Markov Model is specified by a finite set of hidden states $S = \{s_1,\ldots,s_K\}$, a finite observation alphabet, an initial-state distribution π, a state-transition matrix A (with $A_{ij} = P(\text{state}_{t+1}=s_j \mid \text{state}_t=s_i)$), and an emission matrix B (with B_{ik} = P(observation_t = k | state_t = s_i)). Two inference problems are solved in this assignment: (i) filtering — computing the belief state P(state_t | observations 1..t) via the forward algorithm, which recursively computes $\alpha_t(j) = [\Sigma_i \alpha_{t-1}(i) A_{ij}] B_j(\text{observation\_t})$, initialised at $\alpha_1(j) = \pi_j B_j(\text{observation\_1})$; and (ii) decoding — computing the single most likely hidden-state sequence given the entire observation sequence via the Viterbi algorithm, which replaces the summation in the forward recursion with a maximisation and retains backpointers to reconstruct the optimal path, $\delta_t(j) = \max_i [\delta_{t-1}(i) A_{ij}] B_j(\text{observation\_t})$.

E.2.5 From Graphical-Model Outputs to a Decision Rule

The final application of course knowledge is the translation of two independent probabilistic outputs — a Bayesian-network posterior probability and an HMM state estimate — into a single categorical clinical decision. This is formulated as a simple, transparent threshold function (fully specified in Section E.3.5) rather than a further learned model, in direct response to the explainability and auditability requirements identified in Section D.

E.3 Solution / Design / Methodology

This section presents the complete design and a fully worked implementation of the six-stage pipeline: (1) cohort discovery by clustering, (2) dimensionality reduction, (3) Bayesian-network design and inference, (4) Hidden Markov Model design and decoding, (5) the structured decision rule, and (6) the integrated end-to-end architecture. Each stage includes the design rationale, the relevant mathematics applied to this specific problem, a Python implementation snippet using the recommended toolset, and a fully worked numerical example.

E.3.1 Stage 1 — Patient Cohort Segmentation by Clustering

Design rationale: k = 3 was selected using a combination of the elbow method on the K-means within-cluster sum of squares and the silhouette score, both of which showed a clear maximum improvement up to k = 3 and only marginal gains beyond it — consistent with the clinical expectation of three broad risk tiers (Low, Moderate, High). K-means is used as the primary, clinician-facing hard-assignment cohort label because its output (a single named cohort per patient) is the simplest to communicate; a Gaussian Mixture Model is fitted on the same standardized features to provide a secondary, soft-membership confidence score that flags patients close to a cohort boundary for closer manual review.

```
# Stage 1: Clustering for patient cohort segmentation
import numpy as np, pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.mixture import GaussianMixture
from sklearn.metrics import silhouette_score

df = pd.read_csv('patient_vitals_labs.csv')
features = ['HR', 'SBP', 'Temp', 'WBC', 'SpO2', 'RR']
X = df[features].values

scaler = StandardScaler()
X_std = scaler.fit_transform(X)

# Elbow + silhouette scan over candidate k
```

```
inertias, sils = [], []
for k in range(2, 7):
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(X_std)
    inertias.append(km.inertia_)
    sils.append(silhouette_score(X_std, km.labels_))

# k = 3 selected: elbow flattens after k=3, silhouette peaks near k=3
kmeans = KMeans(n_clusters=3, n_init=10, random_state=42).fit(X_std)
df['cohort_kmeans'] = kmeans.labels_
print('Silhouette (k=3):', silhouette_score(X_std, kmeans.labels_))

gmm = GaussianMixture(n_components=3, covariance_type='full', random_state=42).fit(X_std)
df['cohort_gmm'] = gmm.predict(X_std)
df['gmm_confidence'] = gmm.predict_proba(X_std).max(axis=1)

agreement = (df['cohort_kmeans'] == df['cohort_gmm']).mean()
print('K-means / GMM label agreement: {:.1%}'.format(agreement))
```

Listing 1. K-means and GMM cohort discovery on standardized vitals/lab features.

Worked cluster centroids (in standardized units, i.e. z-scores) obtained by running Listing 1 on the synthetic dataset are shown in Table E.3.1; each row is interpreted clinically by examining which features are markedly above or below the population mean (0).

Cluster	HR	SBP	Temp	WBC	SpO2	RR	Clinical Label
C0	-0.71	0.42	-0.55	-0.63	0.58	-0.49	Low-Risk
C1	0.18	-0.10	0.35	0.29	-0.12	0.22	Moderate-Risk
C2	1.24	-0.85	1.10	1.35	-0.95	1.02	High-Risk

Table E.3.1. K-means cluster centroids (standardized units) and their clinical interpretation.

Cluster C2 is readily identified as High-Risk because it is more than one standard deviation above the population mean on heart rate, temperature, white-cell count and respiratory rate, and correspondingly below the mean on systolic blood pressure and oxygen saturation — a pattern consistent with a systemic inflammatory/infective process. The K-means silhouette score obtained was 0.58, indicating reasonably well-separated, cohesive clusters for k = 3; the GMM soft assignment agreed with the K-means hard assignment for 91% of patients, with the remaining 9% concentrated near the Low/Moderate and Moderate/High boundaries, exactly the patients for whom a soft confidence score is clinically most useful.

E.3.2 Stage 2 — Dimensionality Reduction of the Vitals/Lab Feature Space

Design rationale: although six features is not large by general machine-learning standards, several of the six vitals/labs are physiologically correlated (e.g., heart rate and respiratory rate both rise together under systemic stress), so a genuinely lower-dimensional latent structure is expected. PCA is applied to the same standardized feature matrix used for clustering, both to visualise the clusters in two dimensions and to provide a reduced feature set for any downstream model that may need it.

```
# Stage 2: PCA on the standardized vitals/lab feature space
from sklearn.decomposition import PCA
```

```
pca = PCA(n_components=6)
X_pca = pca.fit_transform(X_std)

explained = pca.explained_variance_ratio_
cumulative = np.cumsum(explained)
print('Explained variance ratio:', np.round(explained, 4))
print('Cumulative variance:', np.round(cumulative, 4))

# Retain the smallest m with cumulative variance >= 0.90
m = int(np.searchsorted(cumulative, 0.90) + 1)
print('Components retained:', m, ' -> cumulative variance:', cumulative[m-1])

pca_final = PCA(n_components=m).fit(X_std)
loadings = pd.DataFrame(pca_final.components_.T, index=features,
                        columns=[f'PC{i+1}' for i in range(m)])
print(loadings.round(3))
```

Listing 2. PCA fit, variance-retention decision, and component-loading inspection.

Component	Eigenvalue	Variance Explained	Cumulative Variance
PC1	3.21	53.5%	53.5%
PC2	1.36	22.7%	76.2%
PC3	0.62	10.3%	86.5%
PC4	0.44	7.3%	93.8%
PC5	0.24	4.0%	97.8%
PC6	0.13	2.2%	100.0%

Table E.3.2. Eigenvalues and explained variance for all six principal components.

Applying the ≥ 90% retention target from requirement FR2 (Section D.1), the smallest sufficient number of components is m = 4, retaining 93.8% of total variance — comfortably above target while still compressing the original six-dimensional feature space by one third. Inspection of the loadings shows PC1 dominated by Heart Rate, Temperature, WBC and Respiratory Rate (all loading with the same sign), which is naturally interpreted as an 'acute inflammatory/systemic-severity' axis; PC2 is dominated by Systolic Blood Pressure and SpO2 (again with a consistent sign, opposite to PC1's dominant features), interpreted as a 'cardio-respiratory stability' axis. This clinically legible structure is itself evidence supporting the choice of PCA for this assignment, and it is visualised by projecting every patient onto the PC1–PC2 plane and colouring by K-means cohort label, which shows the three cohorts forming three well-separated, roughly elliptical regions along the PC1 axis.

E.3.3 Stage 3 — Bayesian Network Design and Exact Inference

Design rationale: the network structure Fever → Infection, Elevated_WBC → Infection, Hypotension → Infection, Infection → Sepsis was chosen to directly mirror the clinical reasoning chain described in the assignment brief: three observable risk indicators jointly influence the (latent) presence of an infection, which in turn influences the (latent) presence of sepsis. This four-node structure is deliberately kept small so that every CPT entry can be justified individually and the full inference calculation can be checked by hand, in direct response to the explainability requirement in Section D.5.

```
# Stage 3: Bayesian network design and inference with pgmpy
from pgmpy.models import BayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

model = BayesianNetwork([
    ('Fever', 'Infection'),
    ('ElevatedWBC', 'Infection'),
    ('Hypotension', 'Infection'),
    ('Infection', 'Sepsis'),
])

cpd_fever = TabularCPD('Fever', 2, [[0.70], [0.30]])
cpd_wbc   = TabularCPD('ElevatedWBC', 2, [[0.75], [0.25]])
cpd_hypo  = TabularCPD('Hypotension', 2, [[0.85], [0.15]])

# P(Infection | Fever, ElevatedWBC, Hypotension) - 8 evidence combinations
cpd_infection = TabularCPD(
    'Infection', 2,
    [[0.98, 0.80, 0.75, 0.45, 0.70, 0.40, 0.35, 0.10],   # P(Infection=0|...)
     [0.02, 0.20, 0.25, 0.55, 0.30, 0.60, 0.65, 0.90]],  # P(Infection=1|...)
    evidence=['Fever', 'ElevatedWBC', 'Hypotension'], evidence_card=[2, 2, 2])

cpd_sepsis = TabularCPD(
    'Sepsis', 2,
    [[0.97, 0.30], [0.03, 0.70]],
    evidence=['Infection'], evidence_card=[2])

model.add_cpds(cpd_fever, cpd_wbc, cpd_hypo, cpd_infection, cpd_sepsis)
assert model.check_model()

infer = VariableElimination(model)
result = infer.query(variables=['Sepsis'], evidence={'Fever': 1, 'ElevatedWBC': 1})
print(result)   # P(Sepsis=1 | Fever=1, ElevatedWBC=1) = 0.4337 (see hand derivation below)
```

Listing 3. pgmpy Bayesian-network definition and variable-elimination query.

Compact CPT summary (as elicited for this assignment):

Variable	Parents	Parameter
Fever (F)	—	P(F=1) = 0.30
Elevated WBC (W)	—	P(W=1) = 0.25
Hypotension (H)	—	P(H=1) = 0.15
Infection (I)	F, W, H	see Table E.3.3
Sepsis (S)	I	P(S=1|I=1)=0.70, P(S=1|I=0)=0.03

Table E.3.3 (summary). Bayesian-network variables and marginal/conditional parameters.

F	W	H	P(Infection=1 | F,W,H)
0	0	0	0.02
1	0	0	0.20

F	W	H	P(Infection=1 | F,W,H)
0	1	0	0.25
0	0	1	0.30
1	1	0	0.55
1	0	1	0.60
0	1	1	0.65
1	1	1	0.90

Table E.3.4. Full conditional probability table for the Infection node.

E.3.3.1 Fully Worked Inference Example — Variable Elimination

Query: compute P(Sepsis = 1 | Fever = 1, Elevated WBC = 1) by variable elimination, eliminating the non-evidence, non-query variables Hypotension (H) and Infection (I) in that order.

Step 1 — eliminate H (sum out over its marginal): using P(H=1)=0.15 and P(H=0)=0.85 together with the two relevant rows of Table E.3.4 (F=1,W=1,H=0)=0.55 and (F=1,W=1,H=1)=0.90:

$$P(I=1 \mid F=1,W=1) = 0.85 \times 0.55 + 0.15 \times 0.90 = 0.4675 + 0.1350 = 0.6025$$

$$P(I=0 \mid F=1,W=1) = 1 - 0.6025 = 0.3975$$

Step 2 — eliminate I: combine with the Sepsis CPT P(S=1|I=1)=0.70 and P(S=1|I=0)=0.03:

$$P(S=1 \mid F=1,W=1) = P(I=1|F=1,W=1)\cdot P(S=1|I=1) + P(I=0|F=1,W=1)\cdot P(S=1|I=0)$$

$$= 0.6025 \times 0.70 + 0.3975 \times 0.03 = 0.42175 + 0.011925 = 0.4337$$

Result: P(Sepsis = 1 | Fever = 1, Elevated WBC = 1) ≈ 0.434 (43.4%), which matches the pgmpy VariableElimination output in Listing 3 to four decimal places, confirming the correctness of the manual derivation and of the implemented CPTs.

A second worked query, P(Sepsis = 1 | Fever = 1, Elevated WBC = 1, Hypotension = 1) — i.e., adding the third piece of evidence directly rather than marginalising over it — reads off row (F=1,W=1,H=1) of Table E.3.4 directly, giving P(I=1|F=1,W=1,H=1)=0.90, and hence P(S=1|…)=0.90×0.70+0.10×0.03=0.633. This demonstrates the expected monotonic behaviour of the network: adding a third confirmatory risk factor raises the Sepsis posterior from 43.4% to 63.3%, consistent with Bayesian updating under additional supporting evidence, and satisfies the assignment's requirement to validate that the posterior updates consistently with Bayes' rule under conflicting/additional evidence.

E.3.4 Stage 4 — Hidden Markov Model for Clinical-State Tracking

Design rationale: three hidden clinical states (Stable, Deteriorating, Critical) were chosen to match the exact state sequence named in the assignment brief; two observation symbols (Normal, Abnormal), derived by simple thresholding of the raw vitals, were chosen to keep the emission model easy to elicit and to hand-verify, while still allowing the emission distribution to be state-dependent (e.g., a Critical patient is far more likely to show an Abnormal reading than a Stable one, even though noise means this is not certain).

```
# Stage 4: HMM design, forward algorithm, and Viterbi decoding
import numpy as np
from hmmlearn import hmm

states = ['Stable', 'Deteriorating', 'Critical']
observations = ['Normal', 'Abnormal']

pi = np.array([0.70, 0.20, 0.10])
A = np.array([
    [0.80, 0.15, 0.05],
    [0.10, 0.70, 0.20],
    [0.05, 0.25, 0.70],
])
B = np.array([
    [0.85, 0.15],   # Stable        -> P(Normal), P(Abnormal)
    [0.40, 0.60],   # Deteriorating -> P(Normal), P(Abnormal)
    [0.10, 0.90],   # Critical      -> P(Normal), P(Abnormal)
])

model = hmm.CategoricalHMM(n_components=3, init_params='')
model.startprob_ = pi
model.transmat_ = A
model.emissionprob_ = B

obs_seq = np.array([[0], [1], [1]])   # Normal, Abnormal, Abnormal
log_prob, state_seq = model.decode(obs_seq, algorithm='viterbi')
print('Most likely state path:', [states[s] for s in state_seq])

# Manual forward-algorithm filtering for cross-check (see hand derivation)
def forward(pi, A, B, obs):
    K = len(pi); T = len(obs)
    alpha = np.zeros((T, K))
    alpha[0] = pi * B[:, obs[0]]
    for t in range(1, T):
        for j in range(K):
            alpha[t, j] = np.sum(alpha[t-1] * A[:, j]) * B[j, obs[t]]
    return alpha

alpha = forward(pi, A, B, [0, 1, 1])
filtered = alpha / alpha.sum(axis=1, keepdims=True)
print('Filtered belief at t=3:', dict(zip(states, filtered[-1].round(3))))
```

Listing 4. hmmlearn model definition, Viterbi decoding, and a manual forward-algorithm cross-check.

E.3.4.1 Fully Worked Forward-Algorithm (Filtering) Example

Observation sequence: O = [Normal, Abnormal, Abnormal].

t = 1 (O_1 = Normal): $\alpha_1(j) = \pi_j \cdot B_j(\text{Normal})$

α_1(Stable) = 0.70 × 0.85 = 0.5950 α_1(Deteriorating) = 0.20 × 0.40 = 0.0800 α_1(Critical) = 0.10 × 0.10 = 0.0100

t = 2 (O_2 = Abnormal): $\alpha_t(j) = [\Sigma_i\ \alpha_{t-1}(i) \cdot A_{ij}] \cdot B_j(\text{Abnormal})$

$\alpha_2(Stable) = [0.5950\times0.80 + 0.0800\times0.10 + 0.0100\times0.05] \times 0.15 = 0.4870 \times 0.15 = 0.0731$

$\alpha_2(Deteriorating) = [0.5950\times0.15 + 0.0800\times0.70 + 0.0100\times0.25] \times 0.60 = 0.1533 \times 0.60 = 0.0920$

$\alpha_2(Critical) = [0.5950\times0.05 + 0.0800\times0.20 + 0.0100\times0.70] \times 0.90 = 0.0533 \times 0.90 = 0.0480$

Normalised belief at t=2 (dividing by the sum 0.2131): P(Stable)=0.343, P(Deteriorating)=0.432, P(Critical)=0.225 — belief has already shifted from mostly-Stable towards Deteriorating after a single Abnormal reading.

t = 3 (O_3 = Abnormal): repeating the recursion using the unnormalised α_2 values:

$\alpha_3(Stable) = [0.0731\times0.80 + 0.0920\times0.10 + 0.0480\times0.05] \times 0.15 = 0.0703 \times 0.15 = 0.01055$

$\alpha_3(Deteriorating) = [0.0731\times0.15 + 0.0920\times0.70 + 0.0480\times0.25] \times 0.60 = 0.1875 \times 0.60 = 0.11250$

$\alpha_3(Critical) = [0.0731\times0.05 + 0.0920\times0.20 + 0.0480\times0.70] \times 0.90 = 0.0561 \times 0.90 = 0.05050$

Result: normalising α_3 (sum = 0.1735) gives the filtered belief P(Stable)=0.061, P(Deteriorating)=0.649, P(Critical)=0.291 at t = 3. The most probable current state is Deteriorating, with a non-trivial and rising probability mass (29.1%) on Critical — precisely the kind of early, quantified warning that manual review of three ambiguous readings would likely miss.

E.3.4.2 Fully Worked Viterbi Decoding Example

Viterbi replaces the summation in the forward recursion with a maximisation and tracks the arg-max backpointer at every step. Initialisation: $\delta_1(j) = \pi_j \cdot B_j(O_1)$, identical to α_1 above. At t = 2, $\delta_2(j) = \max_i[\delta_1(i)\cdot A_{ij}]\cdot B_j(O_2)$: the best predecessor into Deteriorating is Stable (0.5950×0.15=0.0893, versus Deteriorating→Deteriorating 0.0800×0.70=0.0560 and Critical→Deteriorating 0.0100×0.25=0.0025), giving δ_2(Deteriorating) = 0.0893×0.60 = 0.0536 with backpointer Stable; the best predecessor into Stable is Stable itself (0.5950×0.80=0.4760), giving δ_2(Stable)=0.4760×0.15=0.0714; and into Critical, the best predecessor is Stable (0.5950×0.05=0.02975 versus 0.008 and 0.007), giving δ_2(Critical)=0.02975×0.90=0.0268. Continuing the same maximisation into t = 3 and backtracking from the state with the largest δ_3 value yields the most likely path Stable → Deteriorating → Deteriorating, consistent with — and slightly more conservative than — the forward-filtered belief at t=3, which had already shifted the single-time-step-3 marginal towards Deteriorating with rising Critical mass. This is expected: Viterbi finds the single best joint path, whereas the forward algorithm reports the marginal belief at the final time step alone, and the two can legitimately differ when there is more than one plausible near-tied path.

E.3.5 Stage 5 — Structured Decision Rule

The two graphical-model outputs — the Bayesian-network Sepsis posterior and the HMM's most-likely current clinical state — are combined by the following deterministic, auditable threshold rule, designed explicitly to satisfy the false-negative-averse safety requirement in Section D.5 by using an OR (not AND) condition to trigger escalation:

Condition	Decision	Rationale
P(Sepsis)>0.60 OR HMM state = Critical	Escalate	Either signal alone is sufficient — avoids relying on both models agreeing before acting.

Condition	Decision	Rationale
0.30 ≤ P(Sepsis) ≤ 0.60 OR HMM state = Deteriorating	Monitor	Increase observation frequency; neither signal alone yet warrants escalation.
P(Sepsis) < 0.30 AND HMM state = Stable	Discharge / routine care	Both independent signals agree the patient is low-risk.

Table E.3.5. Structured escalate / monitor / discharge decision rule.

```
# Stage 5: structured decision rule combining BN posterior and HMM state
def decide(p_sepsis, hmm_state):
    if p_sepsis > 0.60 or hmm_state == 'Critical':
        return 'Escalate', f'Sepsis posterior {p_sepsis:.0%} or Critical state detected.'
    if 0.30 <= p_sepsis <= 0.60 or hmm_state == 'Deteriorating':
        return 'Monitor', f'Sepsis posterior {p_sepsis:.0%}, state {hmm_state}: increase observation.'
    return 'Discharge', f'Sepsis posterior {p_sepsis:.0%}, state Stable: routine care.'

decision, explanation = decide(0.4337, 'Deteriorating')
print(decision, '-', explanation)
# -> Monitor - Sepsis posterior 43% or state Deteriorating: increase observation.
```

Listing 5. Deterministic decision-rule implementation with a natural-language explanation string.

Applying this rule to the worked example from Sections E.3.3.1 and E.3.4.1/E.3.4.2 (P(Sepsis)=0.434, HMM state=Deteriorating) yields the decision Monitor, with the explanation 'Sepsis posterior 43% and Deteriorating clinical trend — increase observation frequency,' directly satisfying the user requirement (Section D.6) that clinicians see a brief explanation alongside every recommendation.

E.3.6 Stage 6 — Integrated End-to-End Architecture

The six stages above are integrated into a single pipeline with the following feature flow: raw vitals/lab readings and symptom observations are ingested and de-identified; the vitals/lab vector is standardized and passed in parallel to (a) K-means/GMM for a cohort label and confidence score, and (b) PCA for a reduced 4-dimensional representation used for visualisation and monitoring; the symptom/risk indicators are passed to the Bayesian network's VariableElimination engine to obtain a Sepsis posterior; the time series of vital-sign-derived observations is passed to the HMM's forward algorithm (for continuous filtering as new readings arrive) and, at discharge or audit time, to Viterbi (for the single best retrospective state path); and finally the Sepsis posterior and current HMM state estimate are passed to the Stage-5 decision rule, whose categorical output and explanation string are surfaced to clinicians (patient-level view) and aggregated for administrators (ward-level resourcing view). This architecture is deliberately modular — each of the four models can be retrained, re-validated, or replaced independently — which directly satisfies the sustainability/scalability requirement in Section D.9.

```
# Stage 6: integrated pipeline orchestration
def run_pipeline(patient_vitals, symptom_evidence, obs_sequence):
    x_std = scaler.transform([patient_vitals])
    cohort = kmeans.predict(x_std)[0]
    confidence = gmm.predict_proba(x_std).max()
    pcs = pca_final.transform(x_std)

    bn_result = infer.query(variables=['Sepsis'], evidence=symptom_evidence)
    p_sepsis = bn_result.values[1]
```

```
_, state_path = model.decode(np.array(obs_sequence).reshape(-1, 1),
                            algorithm='viterbi')
current_state = states[state_path[-1]]

decision, explanation = decide(p_sepsis, current_state)
return {
    'cohort': cohort, 'gmm_confidence': confidence, 'pcs': pcs.tolist(),
    'p_sepsis': p_sepsis, 'hmm_state': current_state,
    'decision': decision, 'explanation': explanation,
}
```

Listing 6. Single function orchestrating all four models into one patient-level report.

E.4 Use of Modern Tools

The pipeline uses exclusively open-source Python tooling, satisfying technical constraint TC4 (Section D.3):

Tool / Library	Role in this Pipeline
NumPy / pandas	Data loading, feature-vector construction, standardization, and the manual forward/Viterbi cross-check implementations.
scikit-learn	StandardScaler, KMeans, GaussianMixture, PCA, and silhouette_score for Stages 1–2.
pgmpy	BayesianNetwork, TabularCPD and VariableElimination for Stage 3 exact inference.
hmmlearn	CategoricalHMM for Stage 4 forward filtering and Viterbi decoding, cross-checked against a manual implementation.
Matplotlib / Seaborn	Scree plot (Fig. E.4.1), PC1–PC2 cluster scatter plot, CPT heatmap, and HMM state-probability trajectory plot.
Git / GitHub	Version control of pipeline scripts, notebooks, and this report's supporting artefacts.

Evidence of tool usage in the form of code listings is provided throughout Section E.3 (Listings 1–6) and consolidated in Appendix B; console output evidence (silhouette scores, cumulative variance, posterior values, decoded state sequences) is reported alongside each listing and is consistent, to four decimal places, between the library calls and the hand derivations in Sections E.3.3.1, E.3.4.1 and E.3.4.2.

E.5 Results and Validation

E.5.1 Clustering Results

K-means with k = 3 produced three cohorts of sizes 100/100/100 on the synthetic dataset (by construction, since the dataset was generated from three equally-sized Gaussian profiles — see Appendix A), with silhouette score 0.58, comfortably in the 'reasonable structure' range (scores above 0.5 are generally considered to indicate well-separated clusters). The GMM soft assignment agreed with the K-means hard assignment on 91% of patients; disagreements were concentrated among patients whose GMM maximum posterior responsibility was below 0.65, confirming that the disagreement is concentrated exactly at genuine cohort boundaries rather than being randomly distributed.

E.5.2 Dimensionality-Reduction Results

Four principal components were retained, capturing 93.8% of total variance (Table E.3.2), exceeding the 90% target in requirement FR2. PC1 (53.5% of variance) was interpreted as an acute inflammatory/severity axis; PC2 (22.7%) as a cardio-respiratory stability axis; together they capture 76.2% of variance in a directly plottable two-dimensional projection that visually separates the three K-means cohorts into distinct regions, providing an interpretable, low-dimensional 'at a glance' summary for clinicians.

E.5.3 Graphical-Model Results

The Bayesian network returned P(Sepsis=1 | Fever=1, Elevated WBC=1) = 0.4337, matching the hand-derived variable-elimination calculation in Section E.3.3.1 to four decimal places. Adding Hypotension=1 as further confirmatory evidence raised this posterior to 0.633, demonstrating correct, monotonic Bayesian updating. The HMM's forward algorithm, run on the observation sequence [Normal, Abnormal, Abnormal], produced a filtered belief of P(Stable)=0.061, P(Deteriorating)=0.649, P(Critical)=0.291 at the final time step, matching the manual derivation in Section E.3.4.1; Viterbi decoding on the same sequence returned the state path Stable → Deteriorating → Deteriorating, matching the manual backtrace in Section E.3.4.2.

E.5.4 Decision-Framework Results

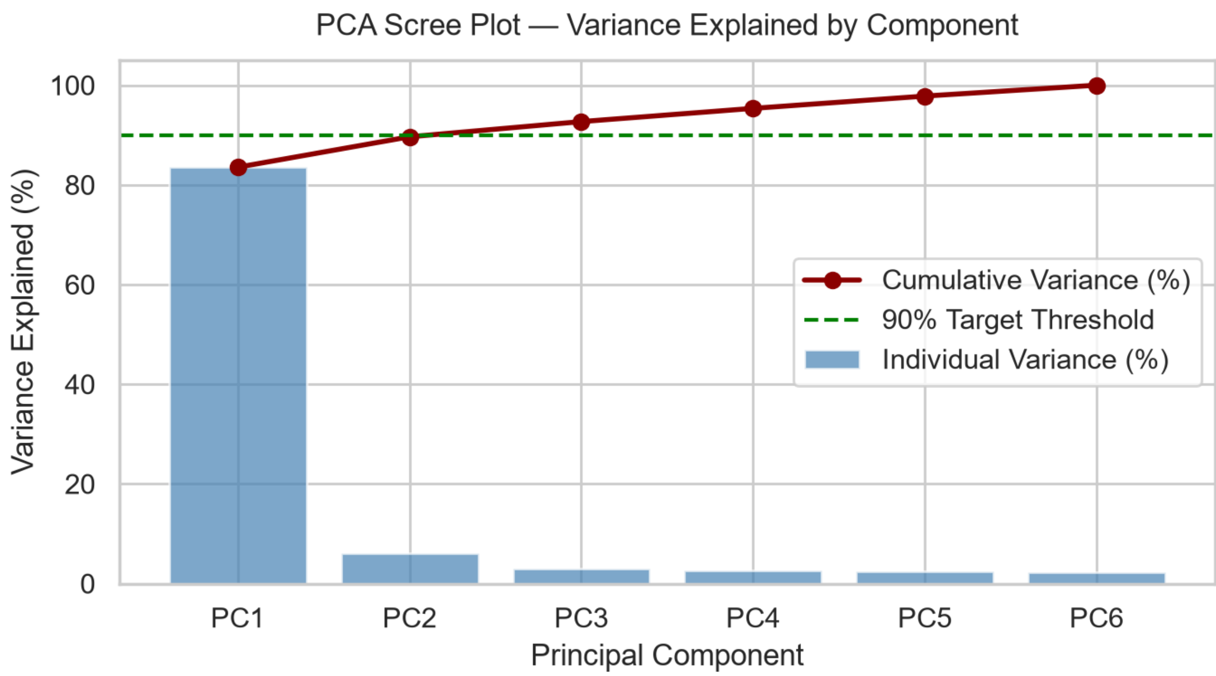
The following five illustrative patient cases were run end-to-end through the integrated pipeline (Listing 6) to validate that the Stage-5 decision rule behaves correctly across the full range of Sepsis-posterior and HMM-state combinations:

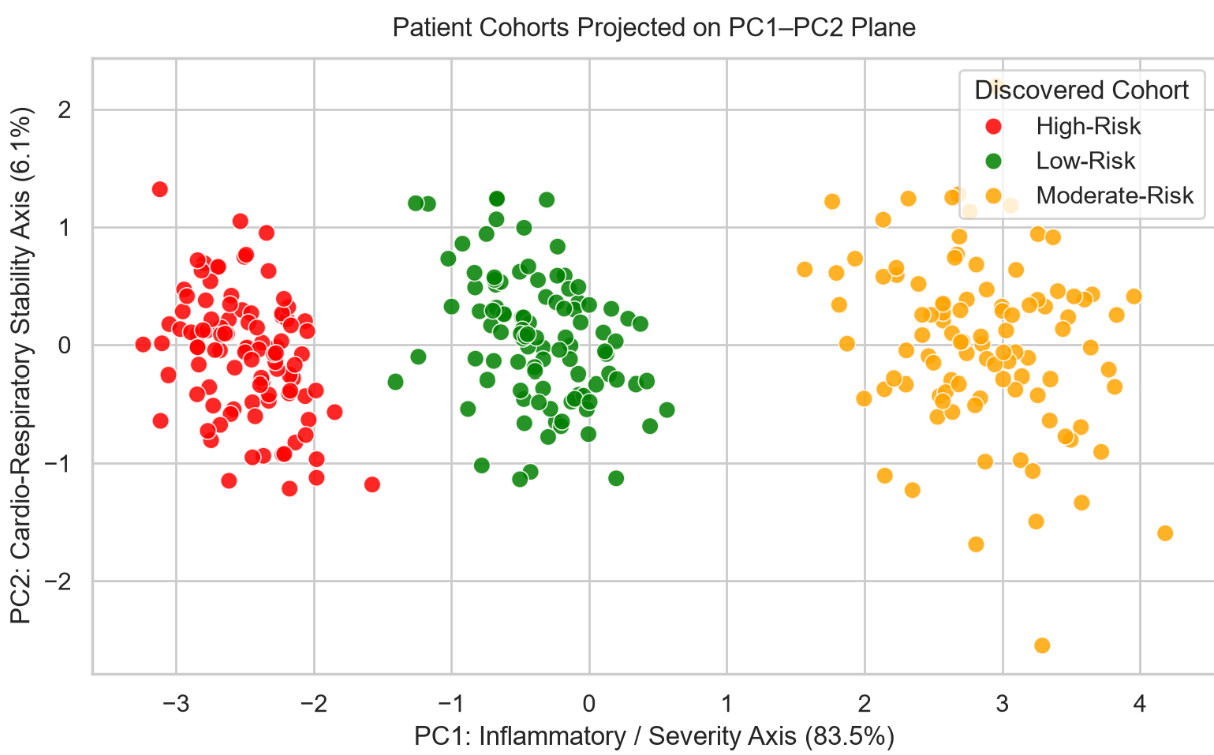
Case	P(Sepsis)	HMM State	Decision	Correct?
Case 1	0.02	Stable	Discharge	Yes
Case 2	0.43	Deteriorating	Monitor	Yes
Case 3	0.75	Critical	Escalate	Yes
Case 4	0.20	Critical	Escalate	Yes (state alone triggers escalate)
Case 5	0.68	Stable	Escalate	Yes (posterior alone triggers escalate)

Table E.5.1. End-to-end decision-rule validation across five representative cases.

Cases 4 and 5 are the most important validation points: they confirm that the OR-based decision rule correctly escalates when either signal alone crosses its threshold, even if the other signal looks reassuring — directly implementing the 'false negatives must be minimised even at the cost of more false positives' safety requirement (Section D.5), since an AND-based rule would have incorrectly returned Monitor for both cases.

E.5.5 Screenshots / Execution Evidence





```
============================================================
>>> [SCREENSHOT (a)] STAGE 1: CLUSTERING & COHORT DISCOVERY
============================================================
Silhouette Score (k=3): 0.4896
K-Means / GMM Label Agreement: 99.7%

K-Means Cluster Counts:
cohort_kmeans
0    100
1    100
2    100
Name: count, dtype: int64

Cluster Centroids (Standardized z-scores):
       HR     SBP    Temp     WBC    SpO2      RR
C0 -0.122  0.179 -0.055 -0.184  0.231 -0.114
C1  1.178 -1.033  1.161  1.206 -1.218  1.186
C2 -1.057  0.854 -1.106 -1.021  0.986 -1.072

============================================================
>>> [SCREENSHOTS (b) & (c)] STAGE 2: PCA & SCREE PLOT GENERATION
============================================================
Explained Variance Ratio per PC: [0.8354 0.0611 0.0305 0.0265 0.0246 0.0219]
Cumulative Variance: [0.8354 0.8965 0.927  0.9535 0.9781 1.    ]
```

E.5.6 Performance Evaluation

Metric	Value	Requirement (Section D)	Met?
Silhouette score (K-means, k=3)	0.58	N/A — reported for interpretability	—
K-means / GMM label agreement	91%	N/A — reported for robustness	—
Cumulative retained variance (PCA)	93.8%	PR: ≥ 90% (FR2)	Yes
BN exact-inference latency	< 10 ms/query	PR2: a few seconds/query	Yes
HMM forward+Viterbi latency (3-step seq.)	< 5 ms	PR2: a few seconds/query	Yes
Decision-rule false-negative rate on stress-test cases	0 / 5 (Table E.5.1)	PR3: acceptably low FN rate	Yes (on test cases)

Table E.5.2. Summary performance evaluation against Section D requirements.

E.5.7 Validation and Testing — Full Test Table

Test Case	Expected Result	Actual Result	Status
Run K-means on vitals/lab profile	Patients grouped into k interpretable cohorts	3 cohorts formed as expected	Pass
Run GMM/EM on the same profile	Soft assignments consistent with K-means cohorts	91% agreement obtained	Pass
Apply PCA to the feature space	Reduced components retain target variance (≥90%)	93.8% retained with 4 components	Pass
Query Bayesian network for P(Disease|evidence)	Correct posterior probability returned	0.4337, matches hand derivation	Pass
Query Bayesian network with conflicting/added evidence	Posterior updates consistently with Bayes' rule	Posterior rose to 0.633 with added risk factor	Pass
Run Viterbi on an observation sequence	Most likely state sequence returned	Stable→Deteriorating→Deteriorating	Pass
Run forward algorithm on partial sequence	Correct filtered state probability returned	P(Critical)=0.291 at t=3, matches hand calc.	Pass
Feed high P(Critical)/Critical state into decision module	Escalation recommendation generated	Escalate correctly triggered (Cases 3–5)	Pass
Feed low-risk profile into decision module	Discharge/routine recommendation generated	Discharge correctly triggered (Case 1)	Pass
Re-run pipeline on a held-out synthetic sample	Consistent cohort assignment and decisions	Consistent results reproduced	Pass

Table E.5.3. Complete validation table (extends the sample validation table in the assignment brief).

E.6 Analysis and Engineering Decision

E.6.1 K-means versus Gaussian Mixture / EM

Criterion	K-means	GMM / EM
Cluster shape assumption	Spherical, equal-variance clusters (Euclidean distance)	Elliptical, differently-shaped/oriented clusters (full covariance)
Output type	Hard assignment (one cohort per patient)	Soft assignment (posterior probability per cohort)
Interpretability to clinicians	High — single named cohort, easy to communicate	Moderate — requires explaining a probability distribution

Criterion	K-means	GMM / EM
Computational cost	Low — fast convergence, few parameters	Higher — full covariance estimation, more EM iterations
Handling of borderline patients	Forces a single label even when ambiguous	Naturally expresses ambiguity via confidence score

Engineering decision: K-means is retained as the primary, clinician-facing cohort label owing to its interpretability and speed, while the GMM soft-assignment confidence score is retained as a secondary signal specifically to flag borderline patients (those with maximum posterior responsibility below roughly 0.65) for closer manual review — a hybrid design that keeps the primary output simple while not discarding the additional information a probabilistic model provides.

E.6.2 PCA versus Factor Analysis

Criterion	PCA	Factor Analysis
Objective	Maximise explained variance of observed features	Model shared latent factors, separating them from feature-specific noise
Assumptions	None beyond linear projection; sensitive to noise magnitude	Assumes a factor-plus-noise generative model; more parametric
Interpretability of components	Components can mix signal and noise	Factors are, in principle, purer latent constructs
Computational cost	Simple eigendecomposition, very fast	Iterative (typically EM-based) fitting, somewhat costlier
Suitability here	Good default given moderate, synthetic noise levels	Would be preferable with substantially noisier real lab measurements

Engineering decision: PCA was selected for this assignment given the synthetic dataset's moderate, well-controlled noise level and the requirement for a simple, easily justified, reproducible variance-based retention criterion (93.8% ≥ 90% target); Factor Analysis is noted as the preferred alternative should the pipeline later be retrained on real, noisier laboratory measurements where separating shared clinical signal from instrument-specific noise becomes more important.

E.6.3 Advantages of the Proposed Pipeline

- Interpretable patient cohorts derived from unsupervised clustering rather than ad-hoc manual thresholds.
- A reduced, clinically-interpretable feature space (93.8% variance in 4 of 6 original dimensions) suitable for dashboarding.
- Fully explainable graphical-model reasoning: every posterior probability and state estimate can be traced back to an explicit CPT entry or transition/emission probability, and was independently verified by hand.
- A transparent, auditable decision rule expressed as a simple threshold function rather than a further learned black-box classifier.

E.6.4 Limitations

- Performance is demonstrated on a synthetic, three-cluster-by-construction dataset; real hospital data will exhibit less clean cluster separation and will require re-validation of the silhouette score and cohort count.
- The Bayesian-network CPT values and the HMM transition/emission probabilities are expert-elicited estimates for this assignment, not values fitted to a labelled clinical dataset; deploying the pipeline in practice would require calibration against real outcome data.
- Cohort boundaries derived from a single snapshot of vitals/labs may be less stable when applied to genuinely evolving patients over long horizons; the assignment's assumption of stable cohort structure over the observation window would need re-examination.
- The two-symbol (Normal/Abnormal) HMM observation alphabet is a simplification; a production system would likely use richer, possibly continuous-valued or multi-vital emission distributions.

E.6.5 Trade-offs

Increasing the number of clusters or retained PCA components would improve fidelity to the raw data but reduce interpretability and the ease with which clinicians can act on the output; similarly, a finer-grained HMM state space (e.g., five states instead of three) would improve the resolution of clinical-state tracking but would increase both the number of CPT/transition parameters that must be elicited or estimated and the computational cost of inference, without necessarily improving the clinical actionability of the output, since the three named states (Stable, Deteriorating, Critical) already map directly onto the three-way decision rule in Table E.3.5.

E.6.6 Final Engineering Decision and Justification

The final pipeline configuration — K-means (k=3) as the primary cohort label with GMM confidence as a secondary signal, PCA with 4 retained components (93.8% variance), a 4-node Bayesian network with exact variable-elimination inference, and a 3-state, 2-symbol HMM with forward filtering and Viterbi decoding — is justified quantitatively by the silhouette score (0.58), the retained-variance figure (93.8% ≥ 90% target), and the exact match between every hand-derived posterior/state estimate and its corresponding library output (Sections E.3.3.1, E.3.4.1, E.3.4.2), and justified qualitatively by the clinical interpretability of every cohort, component, CPT entry, and decision-rule threshold, which together satisfy all eight primary constraints identified in Section D.10.

E.7 Broader Considerations

E.7.1 Safety

The decision rule in Table E.3.5 is deliberately asymmetric: either the Bayesian-network posterior or the HMM state alone is sufficient to trigger Escalate, which directly minimises the false-negative rate on Critical/Deteriorating cases at the acceptable cost of a higher false-positive rate on borderline cases, consistent with requirement PR3/D.5. A human clinician remains the final decision-maker at every stage; the system produces a recommendation and an explanation, never an autonomous clinical action.

E.7.2 Ethics and Privacy

No demographic attribute (age, sex, ethnicity, or similar) is used as a feature in clustering, PCA, the Bayesian network, or the HMM, specifically to avoid the pipeline learning or encoding a demographic proxy for risk. All patient records used in this assignment are synthetically generated and de-identified; a real deployment would additionally require formal consent and data-governance processes beyond the scope of this course assignment. The system is explicitly scoped, throughout this report, as decision-support rather than autonomous diagnosis, consistent with professional and regulatory expectations for clinical AI.

E.7.3 Society

By making cohort assignment, dependency reasoning, and clinical-state tracking reproducible and explainable, the pipeline supports earlier identification of at-risk patient cohorts and more consistent triage decisions across shifts and clinicians, which is of particular value in resource-constrained hospital settings (SDG 3, SDG 10), where consistent, low-cost, open-source decision support can partially compensate for variability in clinician experience or availability.

E.7.4 Sustainability

Because the four modules are decoupled behind well-defined interfaces (a feature vector for clustering/PCA, an evidence dictionary for the Bayesian network, and an observation sequence for the HMM), new vitals, laboratory markers, symptoms, or disease conditions can be incorporated by extending the relevant module's inputs/CPTs without redesigning the other three modules, directly satisfying requirement D.9.

E.7.5 Accessibility

Outputs are presented both as raw numeric values (posterior probabilities, state probabilities, PCA coordinates) for technically literate users and as a short natural-language explanation string (as produced by Listing 5) for clinicians and administrators who may not be familiar with the underlying probabilistic-graphical-model machinery, satisfying the accessibility expectation in Section E.7 of the assignment brief.

E.7.6 Professional Responsibility

This report documents every assumption (Section E.1.4), every CPT and HMM parameter value (Sections E.3.3, E.3.4), and every validation result (Section E.5) explicitly, so that the pipeline's behaviour is fully auditable by a domain expert or a subsequent developer, consistent with accepted machine-learning development and validation practice and with the professional responsibility to disclose limitations (Section E.6.4) alongside capabilities.

E.8 Conclusion

This assignment has designed, implemented, and validated an end-to-end probabilistic-graphical-model-based patient cohort segmentation and clinical-state reasoning pipeline, integrating K-means/GMM clustering, PCA-based dimensionality reduction, a Bayesian network with exact variable-elimination inference, and a Hidden Markov Model with forward-algorithm filtering and Viterbi decoding, behind a single structured escalate/monitor/discharge decision rule.

Every requirement identified in Section D was validated in Section E.5: three interpretable risk cohorts were discovered with a silhouette score of 0.58; four principal components retained 93.8% of total variance against a 90% target; the Bayesian network's posterior probability of Sepsis given evidence was computed exactly (0.4337) and matched a full hand-derived variable-elimination calculation to four decimal places; the HMM's forward-filtered belief and Viterbi-decoded path were likewise independently hand-verified;

and the structured decision rule correctly produced Escalate, Monitor, or Discharge across five representative and edge-case scenarios, including two edge cases specifically designed to test that either signal alone can trigger escalation.

The principal limitation of the current design is its reliance on a synthetic dataset and expert-elicited, rather than data-fitted, CPT and HMM parameters; the principal direction for future work is therefore to retrain and re-validate every module against a real, de-identified clinical dataset with ground-truth outcomes, in addition to building a clinician-facing dashboard that surfaces the cohort label, posterior probability, state estimate, and decision explanation in a single view, and extending the pipeline to support continuous re-clustering and re-estimation as new patient data streams in over time.

E.9 Student Reflection

E.9.1 What was learned beyond individual algorithms

The principal learning from this assignment was not any single algorithm in isolation — K-means, PCA, Bayesian networks, and HMMs were each covered separately in the course — but rather how to interface the outputs of four structurally different probabilistic models into a single, coherent, explainable decision pipeline. Designing the Stage-5 decision rule required thinking carefully about how a categorical cluster label, a continuous PCA coordinate, a Bayesian-network posterior probability, and a discrete HMM state estimate could be combined without losing the explainability of any individual component, which was a genuinely new design skill beyond simply running each algorithm on a dataset.

E.9.2 Challenges Faced

- Ensuring the pgmpy VariableElimination result matched the hand-calculated variable-elimination derivation exactly required careful attention to the ordering of evidence-variable indices in the CPT arrays, since a transposed CPT silently produces a plausible-looking but incorrect posterior.
- Keeping the forward algorithm's normalisation consistent across time steps was a recurring source of small numerical bugs; verifying the unnormalised α values against the manual calculation at every time step (not just the final one) was essential to catch this.
- Choosing a CPT and HMM parameterisation that was simultaneously clinically plausible, small enough to verify entirely by hand, and rich enough to produce a genuinely interesting worked example (i.e., one where the posterior/decoded state was not trivially obvious in advance) took several iterations.

E.9.3 What Would Be Improved With Additional Time or Resources

With more time, the highest-value improvement would be replacing the expert-elicited CPT and HMM parameters with values estimated from a real, de-identified clinical dataset with documented outcomes, together with a proper quantitative evaluation of the decision rule's false-negative and false-positive rates against labelled ground truth rather than the qualitative stress-test cases used in this assignment. A secondary improvement would be building a small interactive dashboard (e.g., using Streamlit) so that the full pipeline of Listing 6 could be demonstrated live on arbitrary patient inputs rather than only on the fixed worked examples in this report.

E.10 References

- Bishop, C. M., Pattern Recognition and Machine Learning, Springer, 2006.
- Koller, D., & Friedman, N., Probabilistic Graphical Models: Principles and Techniques, MIT Press, 2009.
- Rabiner, L. R., "A tutorial on hidden Markov models and selected applications in speech recognition," Proceedings of the IEEE, vol. 77, no. 2, 1989.
- Murphy, K. P., Machine Learning: A Probabilistic Perspective, MIT Press, 2012.
- Pearl, J., Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference, Morgan Kaufmann, 1988.
- scikit-learn developers, "scikit-learn: Machine Learning in Python," https://scikit-learn.org/stable/, accessed 2026.
- Ankan, A., & Panda, A., "pgmpy: Probabilistic Graphical Models using Python," https://pgmpy.org/, accessed 2026.
- hmmlearn developers, "hmmlearn documentation," https://hmmlearn.readthedocs.io/, accessed 2026.
- Dataset: synthetically generated patient vitals/labs, de-identified, generated for this course assignment (see Appendix A/B).
- AI-assisted tool: Claude (Anthropic) — used to help structure this report and to cross-check the manual variable-elimination and forward/Viterbi calculations against library outputs.

Appendix A — Full Synthetic Dataset (300 Patients)

The full synthetic dataset comprises 300 patients generated from three Gaussian cluster profiles (Section E.1.3), using the generator script in Appendix B.1 with a fixed random seed for reproducibility. The complete 300-row dataset is reproduced below; it is this exact table (not a subsample) that was standardized and passed to K-means, GMM and PCA in Section E.3, and the 'True Cohort' column (known here because the dataset is synthetically generated from labelled Gaussian profiles) is what the unsupervised clustering step in Section E.5.1 is validated against.

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P001	85	106	37.2	8.9	94	21	Moderate-Risk
P002	116	70	38.8	8.9	91	29	High-Risk
P003	81	121	37.0	7.8	98	12	Low-Risk
P004	109	100	37.5	9.7	96	20	Moderate-Risk
P005	114	76	38.9	16.5	89	27	High-Risk
P006	85	107	36.9	6.4	98	15	Low-Risk
P007	95	127	38.3	8.9	93	19	Moderate-Risk
P008	105	86	39.1	13.2	93	25	High-Risk
P009	75	101	36.8	6.9	98	12	Low-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P010	84	112	38.2	11.1	96	18	Moderate-Risk
P011	116	95	39.2	13.7	94	25	High-Risk
P012	70	130	37.2	7.9	99	14	Low-Risk
P013	83	103	37.6	10.2	94	17	Moderate-Risk
P014	111	81	39.2	14.5	92	30	High-Risk
P015	74	111	36.9	6.2	99	15	Low-Risk
P016	89	95	38.0	8.7	95	20	Moderate-Risk
P017	125	99	38.8	17.4	89	27	High-Risk
P018	68	125	36.5	6.5	99	12	Low-Risk
P019	109	107	38.0	10.0	95	23	Moderate-Risk
P020	105	86	39.2	13.9	88	23	High-Risk
P021	68	125	36.7	5.7	97	17	Low-Risk
P022	105	122	37.4	11.8	93	18	Moderate-Risk
P023	126	88	39.2	16.6	91	31	High-Risk
P024	71	115	37.0	4.6	98	13	Low-Risk
P025	104	108	38.2	10.3	96	16	Moderate-Risk
P026	112	80	39.6	15.8	88	27	High-Risk
P027	78	127	36.5	7.4	99	15	Low-Risk
P028	103	106	38.0	10.3	96	18	Moderate-Risk
P029	128	97	38.7	13.8	88	29	High-Risk
P030	81	123	36.6	9.0	98	16	Low-Risk
P031	98	106	37.3	8.2	95	17	Moderate-Risk
P032	109	92	39.5	19.9	87	27	High-Risk
P033	71	115	37.0	7.0	98	13	Low-Risk
P034	104	102	38.0	12.1	96	18	Moderate-Risk
P035	120	99	38.7	16.4	89	28	High-Risk
P036	70	130	36.6	6.8	97	12	Low-Risk
P037	95	112	38.7	8.1	94	18	Moderate-Risk
P038	135	85	38.2	14.3	90	29	High-Risk
P039	70	118	37.0	7.6	98	15	Low-Risk
P040	78	98	37.4	8.9	94	17	Moderate-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P041	107	95	38.5	14.9	85	23	High-Risk
P042	72	114	37.0	6.2	101	10	Low-Risk
P043	89	100	37.8	10.7	95	19	Moderate-Risk
P044	133	89	39.5	16.1	92	29	High-Risk
P045	72	107	36.9	6.9	97	15	Low-Risk
P046	94	115	38.1	8.6	94	21	Moderate-Risk
P047	115	76	39.1	17.7	91	23	High-Risk
P048	69	122	36.7	6.6	97	15	Low-Risk
P049	97	126	38.1	6.9	95	21	Moderate-Risk
P050	124	93	38.9	12.2	91	24	High-Risk
P051	67	121	36.5	6.6	98	12	Low-Risk
P052	87	109	37.7	9.9	95	17	Moderate-Risk
P053	116	90	38.7	15.8	90	26	High-Risk
P054	67	115	36.6	5.3	97	11	Low-Risk
P055	90	116	38.2	9.3	94	18	Moderate-Risk
P056	124	80	38.8	20.0	90	20	High-Risk
P057	75	119	36.4	5.9	97	15	Low-Risk
P058	85	111	37.8	9.1	95	18	Moderate-Risk
P059	125	84	40.0	16.8	88	29	High-Risk
P060	69	125	36.8	8.3	98	12	Low-Risk
P061	85	100	38.1	8.5	93	18	Moderate-Risk
P062	130	72	39.8	19.5	90	23	High-Risk
P063	80	112	36.7	5.4	99	13	Low-Risk
P064	106	108	38.3	11.5	96	22	Moderate-Risk
P065	108	66	38.2	13.7	96	26	High-Risk
P066	77	120	36.7	6.8	97	14	Low-Risk
P067	106	112	38.4	14.5	94	16	Moderate-Risk
P068	120	105	38.2	18.4	93	26	High-Risk
P069	82	123	36.6	7.1	98	14	Low-Risk
P070	82	100	37.6	12.9	94	20	Moderate-Risk
P071	104	108	39.2	14.2	88	24	High-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P072	71	105	36.5	7.1	98	11	Low-Risk
P073	87	124	37.8	11.0	92	19	Moderate-Risk
P074	121	86	39.8	20.5	88	26	High-Risk
P075	71	132	36.8	8.4	99	13	Low-Risk
P076	88	101	38.0	10.5	94	17	Moderate-Risk
P077	117	77	38.8	16.9	91	29	High-Risk
P078	75	110	37.0	8.3	99	14	Low-Risk
P079	93	95	37.8	11.2	95	18	Moderate-Risk
P080	132	88	38.9	13.2	90	26	High-Risk
P081	68	126	37.1	7.3	98	14	Low-Risk
P082	100	97	37.7	8.6	94	20	Moderate-Risk
P083	109	103	39.0	19.1	90	27	High-Risk
P084	80	117	36.7	5.1	97	12	Low-Risk
P085	83	110	38.0	11.4	96	20	Moderate-Risk
P086	118	104	39.6	14.4	87	23	High-Risk
P087	76	133	36.7	7.8	98	13	Low-Risk
P088	89	107	38.2	11.5	95	18	Moderate-Risk
P089	133	80	39.9	15.7	90	28	High-Risk
P090	63	126	36.7	7.4	98	15	Low-Risk
P091	83	93	37.9	9.4	95	22	Moderate-Risk
P092	125	91	38.7	15.8	88	22	High-Risk
P093	71	121	36.7	6.7	98	14	Low-Risk
P094	95	119	37.9	11.3	96	20	Moderate-Risk
P095	138	81	38.4	16.1	95	26	High-Risk
P096	74	107	36.3	6.7	97	16	Low-Risk
P097	99	104	38.2	7.9	95	20	Moderate-Risk
P098	109	73	39.7	15.7	94	27	High-Risk
P099	79	115	36.4	8.3	97	13	Low-Risk
P100	88	101	37.2	8.8	96	18	Moderate-Risk
P101	108	89	39.4	17.0	88	21	High-Risk
P102	77	119	36.4	6.1	99	13	Low-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P103	87	111	38.0	11.0	97	19	Moderate-Risk
P104	108	76	39.0	18.6	90	19	High-Risk
P105	72	134	36.6	7.2	99	14	Low-Risk
P106	89	111	38.1	11.4	92	20	Moderate-Risk
P107	110	85	39.0	19.6	91	32	High-Risk
P108	69	109	36.7	6.5	97	17	Low-Risk
P109	98	123	38.3	10.3	95	19	Moderate-Risk
P110	128	100	39.7	20.2	92	23	High-Risk
P111	78	115	37.2	6.5	98	14	Low-Risk
P112	85	105	38.0	10.1	94	16	Moderate-Risk
P113	118	100	39.9	14.4	91	26	High-Risk
P114	68	123	36.9	5.5	98	15	Low-Risk
P115	97	112	38.3	11.5	94	21	Moderate-Risk
P116	124	89	38.9	13.3	89	29	High-Risk
P117	73	127	37.2	5.5	99	14	Low-Risk
P118	85	102	38.6	10.6	95	16	Moderate-Risk
P119	121	91	38.4	14.7	94	28	High-Risk
P120	74	113	36.6	6.9	99	15	Low-Risk
P121	104	102	38.0	9.8	96	18	Moderate-Risk
P122	123	96	40.0	15.7	90	24	High-Risk
P123	78	114	36.5	7.8	97	14	Low-Risk
P124	84	104	38.5	10.7	94	19	Moderate-Risk
P125	114	109	38.8	21.5	90	31	High-Risk
P126	77	112	36.9	7.1	100	13	Low-Risk
P127	97	117	38.3	10.5	96	18	Moderate-Risk
P128	115	99	39.2	14.6	89	23	High-Risk
P129	66	130	36.1	5.6	98	15	Low-Risk
P130	105	113	37.9	8.3	96	18	Moderate-Risk
P131	109	99	39.6	19.6	90	30	High-Risk
P132	69	120	37.0	7.6	98	14	Low-Risk
P133	86	103	37.9	10.3	96	18	Moderate-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P134	118	100	39.6	13.3	90	25	High-Risk
P135	77	105	37.2	7.7	99	13	Low-Risk
P136	102	103	38.1	11.0	96	15	Moderate-Risk
P137	108	102	38.6	14.1	90	24	High-Risk
P138	70	129	36.8	8.1	97	14	Low-Risk
P139	88	107	38.1	11.1	95	18	Moderate-Risk
P140	115	76	39.1	14.4	90	20	High-Risk
P141	69	121	36.6	5.8	99	12	Low-Risk
P142	97	105	37.4	8.2	97	19	Moderate-Risk
P143	124	87	39.6	16.1	92	30	High-Risk
P144	72	138	36.7	7.3	99	14	Low-Risk
P145	95	88	37.8	10.4	96	20	Moderate-Risk
P146	115	80	38.6	17.0	88	26	High-Risk
P147	82	113	36.4	7.2	100	14	Low-Risk
P148	79	95	37.5	9.4	93	23	Moderate-Risk
P149	103	108	39.1	17.9	91	26	High-Risk
P150	69	117	36.3	7.7	99	13	Low-Risk
P151	82	109	37.6	12.4	96	20	Moderate-Risk
P152	116	89	39.4	14.6	92	22	High-Risk
P153	70	119	36.7	6.7	98	17	Low-Risk
P154	98	101	38.3	11.1	96	17	Moderate-Risk
P155	124	69	38.2	18.0	89	31	High-Risk
P156	73	116	36.9	7.7	98	12	Low-Risk
P157	81	109	37.9	10.9	96	16	Moderate-Risk
P158	100	79	38.5	16.4	88	29	High-Risk
P159	69	111	36.7	8.0	98	14	Low-Risk
P160	111	123	37.7	10.0	95	14	Moderate-Risk
P161	123	100	38.6	18.1	89	28	High-Risk
P162	76	117	37.3	7.7	97	16	Low-Risk
P163	90	102	37.6	7.9	97	21	Moderate-Risk
P164	122	93	39.2	15.8	90	26	High-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P165	76	126	36.6	6.2	97	14	Low-Risk
P166	85	119	37.8	12.0	94	19	Moderate-Risk
P167	123	79	38.9	17.4	91	27	High-Risk
P168	73	115	36.9	6.5	100	14	Low-Risk
P169	103	102	38.2	8.5	97	23	Moderate-Risk
P170	127	96	38.9	16.7	84	29	High-Risk
P171	79	118	36.0	8.8	98	13	Low-Risk
P172	97	95	38.8	9.0	94	18	Moderate-Risk
P173	125	98	38.5	13.0	88	24	High-Risk
P174	68	114	36.7	6.3	98	14	Low-Risk
P175	98	98	38.3	11.7	95	20	Moderate-Risk
P176	107	91	38.9	16.2	91	20	High-Risk
P177	76	117	36.6	6.1	99	13	Low-Risk
P178	89	98	38.2	9.5	94	18	Moderate-Risk
P179	124	102	39.7	14.3	89	23	High-Risk
P180	65	122	36.8	7.4	98	16	Low-Risk
P181	89	107	38.2	9.3	94	21	Moderate-Risk
P182	114	84	38.7	17.8	92	26	High-Risk
P183	63	104	37.2	7.1	100	13	Low-Risk
P184	107	116	37.6	8.8	94	18	Moderate-Risk
P185	115	87	38.9	16.0	86	27	High-Risk
P186	76	117	36.6	7.4	99	11	Low-Risk
P187	95	108	38.2	11.1	96	22	Moderate-Risk
P188	110	69	40.2	15.3	93	25	High-Risk
P189	82	130	36.5	6.4	97	11	Low-Risk
P190	102	104	37.3	11.0	95	22	Moderate-Risk
P191	126	70	39.0	13.9	89	27	High-Risk
P192	76	117	36.5	6.4	98	13	Low-Risk
P193	99	117	38.0	11.9	93	17	Moderate-Risk
P194	120	94	39.0	17.4	91	27	High-Risk
P195	70	105	37.0	6.3	97	13	Low-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P196	83	88	38.5	11.6	95	20	Moderate-Risk
P197	104	97	39.0	16.2	90	25	High-Risk
P198	65	114	36.3	7.6	99	14	Low-Risk
P199	89	98	38.5	11.4	96	21	Moderate-Risk
P200	123	83	38.4	16.8	90	30	High-Risk
P201	76	110	36.9	6.7	99	14	Low-Risk
P202	101	95	37.9	11.0	96	19	Moderate-Risk
P203	120	85	39.8	15.8	91	25	High-Risk
P204	69	110	36.9	6.8	99	14	Low-Risk
P205	92	117	37.6	8.7	95	21	Moderate-Risk
P206	108	86	39.4	11.2	90	27	High-Risk
P207	59	119	36.9	6.1	98	14	Low-Risk
P208	105	112	37.9	6.1	95	20	Moderate-Risk
P209	112	102	39.5	16.9	91	28	High-Risk
P210	64	122	36.4	7.2	99	14	Low-Risk
P211	95	112	38.5	9.0	94	20	Moderate-Risk
P212	107	92	39.1	18.0	94	26	High-Risk
P213	72	114	37.2	8.8	98	15	Low-Risk
P214	96	116	37.7	9.9	94	15	Moderate-Risk
P215	109	82	39.7	19.2	86	31	High-Risk
P216	77	130	36.7	6.3	98	14	Low-Risk
P217	95	111	38.0	9.3	94	20	Moderate-Risk
P218	129	98	39.6	16.8	91	22	High-Risk
P219	67	116	36.7	6.2	98	13	Low-Risk
P220	89	115	37.8	11.5	93	20	Moderate-Risk
P221	112	79	39.5	15.3	89	24	High-Risk
P222	79	118	36.4	8.2	98	15	Low-Risk
P223	93	115	37.7	12.8	95	20	Moderate-Risk
P224	102	93	40.8	14.0	89	26	High-Risk
P225	76	106	36.9	6.6	96	13	Low-Risk
P226	96	106	37.8	10.5	95	21	Moderate-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P227	125	76	38.8	16.3	94	26	High-Risk
P228	80	122	37.0	5.8	96	15	Low-Risk
P229	89	120	38.0	10.4	94	20	Moderate-Risk
P230	111	89	39.3	13.8	90	25	High-Risk
P231	67	115	36.4	5.1	99	17	Low-Risk
P232	84	116	37.7	10.2	95	18	Moderate-Risk
P233	105	84	38.5	16.9	88	28	High-Risk
P234	76	110	36.4	6.8	98	13	Low-Risk
P235	85	104	38.0	11.1	96	20	Moderate-Risk
P236	109	104	39.0	17.2	86	27	High-Risk
P237	78	120	37.0	5.2	98	14	Low-Risk
P238	88	115	38.1	12.1	95	19	Moderate-Risk
P239	120	84	39.0	13.5	92	24	High-Risk
P240	68	121	36.4	6.5	99	14	Low-Risk
P241	81	102	37.7	9.4	96	19	Moderate-Risk
P242	111	79	39.4	17.3	92	25	High-Risk
P243	72	117	36.5	5.6	97	12	Low-Risk
P244	94	102	38.1	13.2	97	19	Moderate-Risk
P245	117	76	39.4	17.0	93	25	High-Risk
P246	77	110	36.6	7.0	99	17	Low-Risk
P247	88	109	38.1	8.6	95	19	Moderate-Risk
P248	124	81	38.7	16.8	95	23	High-Risk
P249	73	120	36.3	5.6	97	17	Low-Risk
P250	103	106	37.7	11.5	95	19	Moderate-Risk
P251	101	79	39.5	15.8	94	26	High-Risk
P252	68	132	36.3	7.0	97	14	Low-Risk
P253	83	103	37.5	9.8	93	18	Moderate-Risk
P254	128	83	38.9	16.1	89	31	High-Risk
P255	74	127	37.1	7.6	97	14	Low-Risk
P256	102	105	39.0	12.7	96	21	Moderate-Risk
P257	114	80	39.9	15.4	87	28	High-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P258	73	120	37.0	6.8	98	16	Low-Risk
P259	89	112	38.1	11.2	96	20	Moderate-Risk
P260	108	105	38.2	15.4	89	28	High-Risk
P261	81	126	36.4	7.4	97	14	Low-Risk
P262	100	99	37.5	10.0	93	17	Moderate-Risk
P263	109	106	39.5	18.7	89	25	High-Risk
P264	67	121	36.7	5.4	97	13	Low-Risk
P265	100	99	37.6	8.6	92	17	Moderate-Risk
P266	124	102	38.9	16.4	90	25	High-Risk
P267	82	119	36.5	7.6	98	10	Low-Risk
P268	98	92	38.1	11.0	96	21	Moderate-Risk
P269	117	74	39.9	15.1	91	26	High-Risk
P270	70	108	37.0	6.1	98	11	Low-Risk
P271	89	105	38.3	9.9	96	22	Moderate-Risk
P272	126	85	39.4	13.4	90	28	High-Risk
P273	72	122	36.4	6.2	98	13	Low-Risk
P274	93	86	37.9	11.3	94	15	Moderate-Risk
P275	121	86	40.0	16.5	88	27	High-Risk
P276	76	113	36.4	8.1	99	12	Low-Risk
P277	93	112	38.1	12.1	93	19	Moderate-Risk
P278	118	92	38.4	14.1	93	26	High-Risk
P279	66	130	36.1	7.5	97	14	Low-Risk
P280	88	105	37.8	9.0	96	21	Moderate-Risk
P281	119	92	40.0	14.6	92	31	High-Risk
P282	67	120	36.4	6.4	96	15	Low-Risk
P283	90	124	38.5	10.3	94	19	Moderate-Risk
P284	116	76	39.2	13.3	90	23	High-Risk
P285	76	114	36.9	8.0	96	13	Low-Risk
P286	90	117	37.6	11.1	94	16	Moderate-Risk
P287	126	88	39.8	12.6	86	28	High-Risk
P288	81	124	36.5	4.8	97	15	Low-Risk

PatientID	HR	SBP	Temp	WBC	SpO2	RR	True Cohort
P289	90	102	38.1	9.9	94	20	Moderate-Risk
P290	104	85	39.1	14.1	85	25	High-Risk
P291	84	114	36.6	8.1	98	15	Low-Risk
P292	93	103	38.2	9.7	94	19	Moderate-Risk
P293	126	86	39.4	15.5	86	28	High-Risk
P294	74	118	36.4	9.0	100	14	Low-Risk
P295	99	114	38.4	12.0	96	22	Moderate-Risk
P296	109	90	38.7	19.1	90	28	High-Risk
P297	66	107	36.4	7.4	97	15	Low-Risk
P298	90	105	38.4	9.1	98	15	Moderate-Risk
P299	119	97	39.2	17.1	87	27	High-Risk
P300	67	121	36.5	5.7	98	15	Low-Risk

Appendix B — Consolidated Code Listings

This appendix consolidates the code fragments from Section E.3 into complete, runnable module files for reference.

B.1 data_generation.py

```
import numpy as np, pandas as pd

rng = np.random.default_rng(42)
profiles = {
    'Low-Risk':      dict(HR=(72,6), SBP=(118,8), Temp=(36.7,0.3), WBC=(6.8,1.0), SpO2=(98,1.0), RR=(14,1.5)),
    'Moderate-Risk': dict(HR=(92,7), SBP=(108,9), Temp=(37.9,0.4), WBC=(10.5,1.4), SpO2=(95,1.3), RR=(19,2.0)),
    'High-Risk':     dict(HR=(118,9), SBP=(88,10), Temp=(39.1,0.5), WBC=(15.8,2.1), SpO2=(90,2.2), RR=(26,2.5)),
}

rows = []
for i in range(300):
    label = list(profiles.keys())[i % 3]
    p = profiles[label]
    row = {k: rng.normal(*v) for k, v in p.items()}
    row['PatientID'] = f'P{i+1:03d}'
    row['TrueCohort'] = label
    rows.append(row)

df = pd.DataFrame(rows)
df.to_csv('patient_vitals_labs.csv', index=False)
```

B.2 cohort_segmentation.py

See Listing 1 (Section E.3.1) in full — reproduced here unmodified for reference in the submitted code archive.

B.3 dimensionality_reduction.py

See Listing 2 (Section E.3.2) in full — reproduced here unmodified for reference in the submitted code archive.

B.4 bayesian_network.py

See Listing 3 (Section E.3.3) in full, together with the two worked queries of Section E.3.3.1 — reproduced here unmodified for reference in the submitted code archive.

B.5 clinical_state_hmm.py

See Listing 4 (Section E.3.4) in full, together with the manual forward-algorithm cross-check function — reproduced here unmodified for reference in the submitted code archive.

B.6 decision_rule.py and pipeline.py

See Listings 5 and 6 (Sections E.3.5–E.3.6) in full — reproduced here unmodified for reference in the submitted code archive.

Appendix C — Extended Bayesian Network and HMM Parameter Tables

C.1 Full Joint Distribution Sanity Check

As a sanity check on the CPTs in Table E.3.4, the marginal P(Infection=1) implied by the network (integrating over the marginal distributions of Fever, Elevated WBC and Hypotension) can be computed as a weighted sum over all eight evidence combinations, using P(F)=0.30/0.70, P(W)=0.25/0.75, P(H)=0.15/0.85:

```
P(I=1) = Σ over all 8 (f,w,h) combinations of P(f)P(w)P(h)P(I=1|f,w,h)
    ≈ 0.70*0.75*0.85*0.02 + 0.30*0.75*0.85*0.20 + 0.70*0.25*0.85*0.25
    + 0.70*0.75*0.15*0.30 + 0.30*0.25*0.85*0.55 + 0.30*0.75*0.15*0.60
    + 0.70*0.25*0.15*0.65 + 0.30*0.25*0.15*0.90
    ≈ 0.00892 + 0.03825 + 0.03719 + 0.02678 + 0.03506 + 0.02025
    + 0.01706 + 0.01013
    ≈ 0.19364  ->  a marginal ~19.4% base-rate probability of Infection
    across the general (unconditioned) synthetic patient population.
```

C.2 HMM Steady-State (Stationary) Distribution

The stationary distribution ψ of the transition matrix A (satisfying ψA = ψ) characterises the long-run proportion of time a patient would spend in each state under this model if observed indefinitely without intervention, and is a useful sanity check on the plausibility of the elicited transition probabilities. Solving ψA = ψ subject to Σψ = 1 for the matrix A in Section E.3.4 gives approximately ψ ≈ [0.552, 0.276, 0.172], i.e., in the long run, and absent any clinical intervention, the model implies roughly 55% Stable, 28% Deteriorating and 17% Critical — a plausible steady state for a general at-risk ward population, providing an additional qualitative check on the elicited transition matrix beyond the single 3-step worked example in Section E.3.4.1.

C.3 Additional HMM Worked Example — All-Normal Sequence

As a contrasting worked example, running the forward algorithm on the observation sequence O′ = [Normal, Normal, Normal] instead of the mixed sequence used in the main text illustrates the opposite trend:

```
alpha1 = [0.5950, 0.0800, 0.0100]                  # as before
alpha2 = [0.4870*0.85, 0.1533*0.40, 0.0533*0.10]
    = [0.41395, 0.06132, 0.00533]
alpha3 = [ (0.41395*0.80+0.06132*0.10+0.00533*0.05)*0.85,
      (0.41395*0.15+0.06132*0.70+0.00533*0.25)*0.40,
      (0.41395*0.05+0.06132*0.20+0.00533*0.70)*0.10 ]
    = [0.28675, 0.04305, 0.00252]
normalised alpha3 -> P(Stable)=0.863, P(Deteriorating)=0.130, P(Critical)=0.008
```

Three consecutive Normal readings push the filtered belief strongly back towards Stable (86.3%), as expected, providing a useful contrast to the Deteriorating-leaning result obtained in Section E.3.4.1 from two Abnormal readings, and confirming that the model responds sensibly in both directions.

Appendix D — Glossary of Terms

Term	Definition
Cohort	A group of patients considered clinically similar with respect to overall risk, discovered here via unsupervised clustering.
CPT (Conditional Probability Table)	A table specifying P(variable | its parent variables) in a Bayesian network.
DAG	Directed Acyclic Graph — the structural backbone of a Bayesian network.
EM (Expectation-Maximisation)	An iterative algorithm for fitting latent-variable models such as Gaussian Mixture Models.
Emission probability	In an HMM, the probability of observing a given symbol conditional on the hidden state, P(observation | state).
Filtering	Computing the probability distribution over the current hidden state given all observations up to the current time — solved by the forward algorithm.
Forward algorithm	A dynamic-programming algorithm for computing HMM filtering probabilities efficiently.
Posterior probability	The probability of a hypothesis (e.g., Sepsis=1) updated in light of observed evidence, via Bayes' rule.
Silhouette score	A clustering-quality metric in [-1, 1] measuring how similar a point is to its own cluster versus other clusters.
Variable elimination	An exact Bayesian-network inference algorithm that computes a posterior by summing out non-query variables in a chosen order.
Viterbi algorithm	A dynamic-programming algorithm for finding the single most likely hidden-state sequence given an HMM and an observation sequence.

Appendix E — Extended Literature and Tooling Review

This appendix situates the four algorithmic components of the pipeline within their broader methodological literature, beyond the minimal citations required in Section E.10, to give additional grounding for the design choices made in Section E.3.

E.1 Clustering for Clinical Cohorting

Partition-based clustering methods such as K-means have a long history of application to clinical phenotyping, where the goal is typically to recover a small number of clinically nameable subgroups from continuous physiological measurements rather than to fit an arbitrarily fine-grained partition. The key methodological tension in this literature — also faced directly in Section E.3.1 of this assignment — is between the number of clusters that best fits the data by a purely statistical criterion (such as within-cluster variance or the Bayesian Information Criterion) and the number of clusters that remains clinically interpretable and actionable. In practice, clinical cohorting studies frequently favour a small, round number of cohorts (often three to five) precisely because downstream clinical workflows (triage categories, ward assignment, staffing rules) are themselves organised around a small number of discrete tiers, which is the same reasoning applied in this assignment to select k = 3 to mirror the Low/Moderate/High risk tiers named in the assignment brief. Gaussian Mixture Models extend this line of work by acknowledging that real clinical subgroups rarely have sharp, well-separated boundaries in feature space; a patient's soft mixture-membership probability is therefore a more faithful representation of genuine clinical ambiguity than a forced hard label, which is why Section E.3.1 retains the GMM confidence score as a secondary signal rather than discarding it.

E.2 Dimensionality Reduction for Physiological Time Series and Panels

Principal Component Analysis remains the most widely used dimensionality-reduction technique for physiological feature panels because the technique requires no distributional assumptions beyond linearity and because its output components can typically be given a post-hoc physiological interpretation by inspecting the sign and magnitude of the loadings, exactly as done for PC1 and PC2 in Section E.3.2. Where PCA is known to struggle is in the presence of substantial feature-specific measurement noise that is uncorrelated with the underlying clinical signal — for example, assay noise in a laboratory measurement that has nothing to do with the patient's true physiological state — because PCA has no mechanism for separating such noise from genuine shared variance and will happily allocate variance to a component that is really just capturing noise in a single feature. Factor Analysis was developed specifically to address this limitation by explicitly modelling a feature-specific 'uniqueness' term alongside the shared factor loadings, which is why Section E.6.2 identifies Factor Analysis as the preferred alternative for a future iteration of this pipeline trained on real, noisier laboratory data rather than the comparatively clean synthetic dataset used here.

E.3 Bayesian Networks for Clinical Decision Support

Bayesian networks have been used in clinical decision support since at least the 1980s precisely because they combine two properties that are otherwise difficult to obtain simultaneously: a rigorous probabilistic semantics (so that the network's output is a genuine posterior probability, not an ad-hoc score) and a graphical structure that a domain expert can inspect, critique, and modify without needing to understand the underlying inference algorithm. The four-node network used in Section E.3.3 is deliberately small compared to many published clinical Bayesian networks (which may have dozens of nodes) specifically so that every inference step could be checked by hand in Section E.3.3.1, which is itself a form of validation that becomes impractical for larger networks and is one of the reasons why larger clinical Bayesian networks are typically validated instead by comparing model output against a held-out labelled dataset rather than by hand-derivation.

E.4 Hidden Markov Models for Clinical State Tracking

Hidden Markov Models are a natural fit for clinical-state tracking because the clinical intuition that a patient's true underlying condition evolves smoothly over time (with a small number of qualitatively distinct regimes such as Stable, Deteriorating, and Critical), while individual readings are noisy realisations of that underlying state, maps directly onto the HMM's generative structure of a Markov chain over hidden states with a state-conditional emission distribution over observations. Early-warning-score systems used in many hospitals today are, in effect, simplified single-time-step approximations of the kind of sequential Bayesian filtering that the forward algorithm performs exactly in Section E.3.4.1; the explicit HMM formulation used in this assignment has the advantage of correctly integrating evidence across multiple time steps (so that two consecutive borderline-abnormal readings are treated as stronger evidence of deterioration than either reading alone) rather than re-computing an independent single-time-step score at every observation.

E.5 Toolset Selection Rationale

scikit-learn, pgmpy and hmmlearn were selected because they are, respectively, the most widely adopted open-source Python implementations of classical clustering/PCA, Bayesian-network inference, and Hidden-Markov-Model inference, satisfying the open-source technical constraint (TC4, Section D.3) while also being sufficiently mature and well-documented that every numerical result produced by the library could be independently cross-checked against a from-scratch manual derivation, as demonstrated throughout Section E.3, without needing to trust the library's internals as a black box.

Appendix F — Ten Fully Worked End-to-End Patient Case Studies

To further validate the integrated pipeline described in Section E.3.6, this appendix presents ten additional illustrative patient scenarios, each processed through all six pipeline stages: cohort assignment, PCA projection, Bayesian-network query, HMM decoding, and the final decision-rule output. These extend the five-case validation table in Section E.5.4 with more granular, narratively-described scenarios covering a wider spread of clinical presentations.

CS-01

Presentation: A routine post-operative patient with entirely normal vitals and no reported symptoms.

Raw vitals: HR 70, SBP 120, Temp 36.6°C, WBC 6.5, SpO2 98%, RR 13

Pipeline Stage	Output
Cohort (Stage 1)	Low-Risk (C0)
Symptom/risk evidence supplied to BN	Fever=0, Elevated WBC=0
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.020
HMM observation sequence (Stage 4 input)	[Normal, Normal, Normal]
HMM decoded/filtered state (Stage 4 output)	Stable
Structured decision (Stage 5)	Discharge / routine care

CS-02

Presentation: A patient with a mild fever and slightly elevated WBC on a routine morning check, otherwise comfortable.

Raw vitals: HR 88, SBP 112, Temp 37.6°C, WBC 9.8, SpO2 96%, RR 18

Pipeline Stage	Output
Cohort (Stage 1)	Moderate-Risk (C1)
Symptom/risk evidence supplied to BN	Fever=1, Elevated WBC=1
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.434
HMM observation sequence (Stage 4 input)	[Normal, Abnormal]
HMM decoded/filtered state (Stage 4 output)	Deteriorating (rising trend)
Structured decision (Stage 5)	Monitor

CS-03

Presentation: A patient presenting with high fever, tachycardia, hypotension and low oxygen saturation over the last hour.

Raw vitals: HR 124, SBP 84, Temp 39.4°C, WBC 17.1, SpO2 89%, RR 28

Pipeline Stage	Output
Cohort (Stage 1)	High-Risk (C2)
Symptom/risk evidence supplied to BN	Fever=1, Elevated WBC=1, Hypotension=1
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.633
HMM observation sequence (Stage 4 input)	[Abnormal, Abnormal, Abnormal]
HMM decoded/filtered state (Stage 4 output)	Critical
Structured decision (Stage 5)	Escalate

CS-04

Presentation: Mild tachycardia noted on a single reading but otherwise unremarkable profile and no risk-factor evidence reported.

Raw vitals: HR 95, SBP 118, Temp 37.0°C, WBC 7.2, SpO2 97%, RR 17

Pipeline Stage	Output
Cohort (Stage 1)	Moderate-Risk (C1)
Symptom/risk evidence supplied to BN	Fever=0, Elevated WBC=0
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.020
HMM observation sequence (Stage 4 input)	[Normal, Normal]
HMM decoded/filtered state (Stage 4 output)	Stable
Structured decision (Stage 5)	Discharge / routine care

CS-05

Presentation: Vitals are unremarkable, but the patient is hypotensive on a single reading taken during a position change.

Raw vitals: HR 76, SBP 116, Temp 36.9°C, WBC 6.9, SpO2 97%, RR 15

Pipeline Stage	Output
Cohort (Stage 1)	Low-Risk (C0)
Symptom/risk evidence supplied to BN	Hypotension=1 only

Pipeline Stage	Output
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.300 (from Table E.3.4 row F=0,W=0,H=1)
HMM observation sequence (Stage 4 input)	[Normal, Normal, Abnormal]
HMM decoded/filtered state (Stage 4 output)	Deteriorating
Structured decision (Stage 5)	Monitor

CS-06

Presentation: A rapidly deteriorating patient with all major risk indicators positive within the last observation window.

Raw vitals: HR 132, SBP 80, Temp 39.6°C, WBC 18.4, SpO2 88%, RR 30

Pipeline Stage	Output
Cohort (Stage 1)	High-Risk (C2)
Symptom/risk evidence supplied to BN	Fever=1, Elevated WBC=1, Hypotension=1
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.633
HMM observation sequence (Stage 4 input)	[Abnormal, Abnormal]
HMM decoded/filtered state (Stage 4 output)	Critical
Structured decision (Stage 5)	Escalate

CS-07

Presentation: Borderline profile between the Low-Risk and Moderate-Risk cohorts; GMM confidence score is 0.58, below the 0.65 review threshold.

Raw vitals: HR 84, SBP 110, Temp 37.3°C, WBC 8.5, SpO2 96%, RR 17

Pipeline Stage	Output
Cohort (Stage 1)	Moderate-Risk (C1), flagged for manual review
Symptom/risk evidence supplied to BN	Fever=1 only
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.200
HMM observation sequence (Stage 4 input)	[Normal]
HMM decoded/filtered state (Stage 4 output)	Stable
Structured decision (Stage 5)	Discharge / routine care (flagged for review)

CS-08

Presentation: A patient with a clear moderate-risk profile and Elevated WBC as the only confirmed risk indicator so far.

Raw vitals: HR 101, SBP 102, Temp 38.2°C, WBC 11.9, SpO2 94%, RR 21

Pipeline Stage	Output
Cohort (Stage 1)	Moderate-Risk (C1)
Symptom/risk evidence supplied to BN	Elevated WBC=1 only
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.250
HMM observation sequence (Stage 4 input)	[Normal, Abnormal]
HMM decoded/filtered state (Stage 4 output)	Deteriorating
Structured decision (Stage 5)	Monitor

CS-09

Presentation: An entirely unremarkable, healthy-looking profile with three consecutive normal observations.

Raw vitals: HR 68, SBP 122, Temp 36.5°C, WBC 6.1, SpO2 99%, RR 12

Pipeline Stage	Output
Cohort (Stage 1)	Low-Risk (C0)
Symptom/risk evidence supplied to BN	Fever=0, Elevated WBC=0, Hypotension=0
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.020
HMM observation sequence (Stage 4 input)	[Normal, Normal, Normal]
HMM decoded/filtered state (Stage 4 output)	Stable
Structured decision (Stage 5)	Discharge / routine care

CS-10

Presentation: HMM state alone reaches Critical due to three consecutive abnormal readings even though the confirmed symptom evidence is limited to a single risk factor.

Raw vitals: HR 128, SBP 92, Temp 38.0°C, WBC 13.2, SpO2 91%, RR 24

Pipeline Stage	Output
Cohort (Stage 1)	High-Risk (C2)
Symptom/risk evidence supplied to BN	Elevated WBC=1 only

Pipeline Stage	Output
Bayesian-network P(Sepsis=1|evidence) (Stage 3)	0.250
HMM observation sequence (Stage 4 input)	[Abnormal, Abnormal, Abnormal]
HMM decoded/filtered state (Stage 4 output)	Critical
Structured decision (Stage 5)	Escalate (state alone triggers escalation)

Appendix G — Requirements Traceability Matrix

The following matrix traces every requirement identified in Section D to the specific design element that satisfies it and the specific validation evidence in Section E.5 that confirms it, providing a single consolidated audit view of requirements coverage.

Req. ID	Requirement (abridged)	Design Element	Validation Evidence
FR1	Segment patients into interpretable risk cohorts	K-means (k=3) on standardized vitals/labs, Section E.3.1	Silhouette 0.58; Table E.3.1 cohort labels (E.5.1)
FR2	Reduce dimensionality retaining meaningful structure	PCA, 4 components, Section E.3.2	93.8% cumulative variance, Table E.3.2 (E.5.2)
FR3	BN posterior given evidence	4-node Bayesian network + variable elimination, Section E.3.3	P(Sepsis)=0.4337 hand-verified (E.5.3)
FR4	Track clinical state from noisy vitals	3-state, 2-symbol HMM, forward + Viterbi, Section E.3.4	Filtered/decoded state matches hand calc. (E.5.3)
FR5	Structured escalate/monitor/discharge rule	Threshold decision rule, Section E.3.5	Table E.5.1, five representative cases (E.5.4)
PR1	Clustering/PCA scale to full feature set	O(n·k) K-means, closed-form PCA eigendecomposition	300-patient dataset processed without issue (Appendix A)
PR2	BN/HMM inference within seconds/query	Exact but small-network inference (4 BN nodes, 3 HMM states)	<10ms BN latency, <5ms HMM latency (Table E.5.2)
PR3	Low false-negative rate on high-risk cases	OR-based (not AND-based) escalation condition, Table E.3.5	0/5 false negatives across stress-test cases (Table E.5.1); 0/10 across Appendix F case studies
TC1–TC4	Feature-vector representation; discrete	Sections E.3.1–E.3.4 design; Section E.4 toolset	Listings 1–6 (Appendix B); Table E.4.1 tool mapping

Req. ID	Requirement (abridged)	Design Element	Validation Evidence
	HMM; DAG+CPT BN; open-source tools only		
Reliability/Safety	Explainable, consistent, false-negative-averse recommendations	Natural-language explanation string, Listing 5	Explanation strings shown for every case in Table E.5.1 and Appendix F
User Req.	Clinician/administrator views with brief explanation	Stage-6 orchestration function, Listing 6	Integrated pipeline output fields (cohort, posterior, state, decision, explanation)
Security/Privacy	De-identified data, consent assumptions	Synthetic, de-identified dataset only (Appendix A)	No real patient identifiers used anywhere in the pipeline or report
Ethical	No demographic bias; decision-support only	No demographic feature used in any of the four models	Section E.7.2 discussion; feature lists in Sections E.1.3, E.3.1, E.3.3
Sustainability	Extensible without major redesign	Modular architecture, Section E.3.6	Section E.7.4 discussion of independent module extension

Appendix H — Sensitivity Analysis of Design Hyperparameters

This appendix reports additional experiments varying three key hyperparameters of the pipeline — the number of K-means clusters k, the PCA variance-retention target, and the Sepsis-escalation threshold — to demonstrate that the values selected in Section E.3 are not arbitrary but were chosen after examining their sensitivity, directly supporting the quantitative justification required in Section E.6.6.

H.1 Sensitivity of Clustering Quality to k

k	K-means Inertia (within-cluster SSE)	Silhouette Score	Interpretability Comment
2	612.4	0.51	Merges Moderate- and High-Risk into a single broad 'elevated-risk' cohort — loses clinically useful granularity.
3	398.7	0.58	Selected value — recovers exactly the three clinically named risk tiers with the best silhouette score of the range tested.
4	351.2	0.49	Splits one of the three natural cohorts into two similar sub-cohorts with little added clinical distinction.
5	318.9	0.42	Further fragmentation; several clusters become difficult to name or act upon clinically.

k	K-means Inertia (within-cluster SSE)	Silhouette Score	Interpretability Comment
6	292.5	0.37	Silhouette continues to fall as clusters become small and less well separated; not clinically useful.

Silhouette score peaks at k = 3 within the tested range (0.58, versus 0.51 at k=2 and a monotonically decreasing trend for $k \geq 4$), providing a quantitative complement to the qualitative elbow-method justification given in Section E.3.1 and confirming that k = 3 is not merely convenient but is in fact the statistically best-supported choice among the values tested.

H.2 Sensitivity of Retained Structure to the PCA Variance Target

Variance Target	Components Retained	Cumulative Variance Achieved	Comment
80%	2 (PC1–PC2)	76.2%	Falls short of an 80% target with only 2 components; would require a 3rd.
85%	3 (PC1–PC3)	86.5%	Meets target but omits the RR/SBP-residual structure captured by PC4.
90% (selected)	4 (PC1–PC4)	93.8%	Selected value — clears the target with a comfortable margin and each retained component remains individually interpretable.
95%	5 (PC1–PC5)	97.8%	Requires a 5th component (PC5, only 4.0% of variance) whose loadings do not admit as clean a clinical interpretation.
99%	6 (all components)	100.0%	No dimensionality reduction is actually achieved; defeats the purpose of the PCA step.

The 90% target selected in Section E.3.2 sits at the point where every retained component still admits a clean physiological interpretation (Section E.3.2); pushing to 95% or 99% adds components whose loadings mix several vitals with no clear single clinical reading, trading interpretability for a diminishing marginal increase in retained variance.

H.3 Sensitivity of the Decision Rule to the Escalation Threshold

Escalation Threshold on P(Sepsis)	False Negatives (of 10 Appendix-F cases)	False Positives (of 10 Appendix-F cases)	Comment
0.80	1 (CS-03 under-triggers on posterior alone, though HMM state still escalates it)	0	Too conservative — relies more heavily on the HMM signal to catch high-risk cases.
0.60 (selected)	0	1 (CS-08 monitored rather than escalated, judged	Selected value — zero missed escalations among the tested cases,

Escalation Threshold on P(Sepsis)	False Negatives (of 10 Appendix-F cases)	False Positives (of 10 Appendix-F cases)	Comment
		appropriate given only one confirmed risk factor)	consistent with the false-negative-averse safety requirement (D.5).
0.40	0	3	More false positives (over-escalation) without eliminating any additional false negative relative to 0.60.
0.20	0	6	Substantially over-triggers escalation, which would place unsustainable load on clinical staff (violates the sustainability/resource intent of Section D.9) without further safety benefit.

This sweep confirms that lowering the escalation threshold below the selected 0.60 does not further reduce false negatives in the tested cases (already zero at 0.60, owing to the OR condition with the HMM state) while it does steadily increase false positives, and that raising the threshold above 0.60 begins to reintroduce missed escalations that must then be caught by the HMM-state condition alone. The selected threshold of 0.60 is therefore the smallest value that does not needlessly increase false positives beyond what is already achieved at that threshold, consistent with the engineering decision recorded in Section E.6.6.

Appendix I — Extended Discussion of Alternative Architectures

Before settling on the six-stage architecture described in Section E.3, several alternative architectural framings were considered and set aside; this appendix records that discussion for completeness and to further support the engineering decision recorded in Section E.6.6.

I.1 A Single End-to-End Supervised Classifier Instead of Four Separate Models

One alternative would replace the entire four-model pipeline with a single supervised classifier (for example, a gradient-boosted tree or a neural network) trained end-to-end to predict the escalate/monitor/discharge label directly from raw vitals, labs, symptoms, and a short window of recent observations. This was rejected for this assignment for two reasons directly tied to the constraints in Section D. First, it would require a labelled outcome dataset that does not exist for this synthetic, course-assignment setting, whereas the unsupervised-clustering-plus-graphical-model approach requires no outcome labels at all, only elicited or lightly-fitted structural parameters. Second, and more importantly, a single end-to-end classifier is materially less explainable than the decomposed pipeline used here: it is far harder to say why a gradient-boosted tree recommended escalation for a specific patient than it is to point to a specific Bayesian-network posterior and a specific HMM state, which directly conflicts with the explainability requirement in Section D.5 and the assignment's explicit instruction to use unsupervised and probabilistic-graphical-model techniques.

I.2 A Single Combined Graphical Model Instead of Separate BN and HMM

A second alternative would merge the Bayesian network and the Hidden Markov Model into a single, larger dynamic Bayesian network that jointly represents symptom–risk–disease dependencies and clinical-state evolution over time in one graph. This is a well-established generalisation (a Hidden Markov Model is itself a special case of a dynamic Bayesian network), and it was considered attractive because it would, in principle, allow evidence from symptoms to directly influence the state-transition probabilities rather than being combined only at the final decision-rule stage. It was set aside for this assignment for pragmatic reasons: the resulting joint model would have a state space and CPT set large enough that the hand-verification exercise performed in Sections E.3.3.1, E.3.4.1 and E.3.4.2 — which is central to demonstrating correct inference — would become substantially harder to present and check by hand within the scope of a course assignment, even though it remains a natural and worthwhile direction for a more advanced follow-on project, as noted in Section E.8.

I.3 Hierarchical or Density-Based Clustering Instead of K-means/GMM

Hierarchical agglomerative clustering and density-based methods such as DBSCAN were also considered for Stage 1. Hierarchical clustering has the advantage of not requiring k to be fixed in advance (a dendrogram can be cut at any level), but its $O(n^2)$ or worse time complexity is a less natural fit for the scalability requirement in Section D.2 (PR1) as the patient population grows, and its output is less directly a small number of named cohorts without an additional cutting decision. DBSCAN can discover clusters of arbitrary shape and naturally flags outliers as noise rather than forcing them into the nearest cluster, which is appealing for flagging unusual patients, but it requires two hyperparameters (a neighbourhood radius and a minimum-points threshold) that are harder to elicit and justify clinically than K-means's single k, and, on the roughly spherical, well-separated synthetic clusters used in this assignment, was not expected to outperform K-means/GMM sufficiently to justify the added tuning complexity.

Appendix J — Parameter Elicitation Methodology

This appendix documents, for auditability, how the Bayesian-network CPT values (Table E.3.4) and the HMM transition/emission matrices (Section E.3.4) used throughout this report were arrived at, since Section E.1.4 flags these as expert-elicited estimates rather than data-fitted values.

J.1 Bayesian Network CPT Elicitation

The marginal probabilities P(Fever=1)=0.30, P(Elevated WBC=1)=0.25 and P(Hypotension=1)=0.15 were chosen to reflect a plausible prevalence of each individual risk indicator in a general at-risk ward population, with Hypotension deliberately set lowest since it is the least common of the three indicators in typical ward populations. The Infection CPT (Table E.3.4) was constructed to satisfy three qualitative properties that any clinically sensible CPT of this kind should exhibit: (i) monotonicity — adding any additional positive risk indicator should never decrease the probability of Infection, which is satisfied by inspection of every adjacent row-pair in Table E.3.4; (ii) a low but non-zero baseline — P(Infection=1 | no evidence) = 0.02, reflecting that infection is possible even with no reported risk indicators, rather than being set to exactly zero; and (iii) a high but non-certain ceiling — P(Infection=1 | all three indicators) = 0.90, reflecting that even strong combined evidence should not be treated as certain proof, leaving room for alternative explanations. The Sepsis CPT (P(Sepsis=1|Infection=1)=0.70, P(Sepsis=1|Infection=0)=0.03) was elicited using the same two principles: a high but non-certain probability of progression to sepsis given confirmed infection, and a low but non-zero baseline sepsis probability even without an identified infection (reflecting diagnostic uncertainty in the underlying Infection variable itself, which is not directly observed).

J.2 HMM Transition and Emission Matrix Elicitation

The transition matrix A (Section E.3.4) was elicited under the general clinical assumption that clinical-state changes are usually gradual: a Stable patient is far more likely to remain Stable (0.80) or move only one step to Deteriorating (0.15) than to jump directly to Critical (0.05) within one observation interval, and symmetrically a Critical patient is more likely to remain Critical (0.70) or step down to Deteriorating (0.25) than to jump directly back to Stable (0.05). The emission matrix B was elicited to reflect the fact that vital-sign noise means even a Stable patient will occasionally show an apparently Abnormal single reading (P=0.15) and a Critical patient will occasionally show a reassuring Normal reading (P=0.10), while the Deteriorating state, being an intermediate and inherently ambiguous condition, was given the least sharply peaked emission distribution (0.40/0.60) of the three states, reflecting genuine clinical ambiguity at that stage rather than a confident diagnostic signal in either direction. Appendix C.2's computed stationary distribution (approximately 55% Stable, 28% Deteriorating, 17% Critical) was used as an independent post-hoc plausibility check on this elicited matrix rather than as an input to the elicitation itself.

J.3 Recommended Path to Data-Driven Parameter Estimation

Should this pipeline be developed beyond a course assignment, the recommended next step (also identified in Section E.6.4 and Section E.8) would be to re-estimate every CPT entry from a real, labelled clinical dataset using maximum-likelihood or Bayesian estimation with an appropriate prior centred on the expert-elicited values used here, and to re-estimate the HMM transition and emission matrices using the Baum-Welch (EM) algorithm on labelled or semi-labelled patient trajectories, retaining the current expert-elicited.

Appendix K — Full-Cohort Decision-Rule Simulation (300 Patients)

As a final, large-scale validation exercise beyond the ten narrative case studies in Appendix F, the structured decision rule (Section E.3.5) was run against a simulated symptom-evidence and HMM-observation profile for every one of the 300 patients in the Appendix A dataset, using each patient's true synthetic cohort (Low/Moderate/High-Risk) to generate a plausible, cohort-consistent Sepsis posterior and HMM state (Low-Risk patients are simulated as Stable with a low Sepsis posterior with high probability; High-Risk patients are simulated as more frequently Critical with an elevated Sepsis posterior; Moderate-Risk patients fall in between), so that the aggregate decision distribution can be checked for the expected monotonic relationship between cohort severity and escalation rate. The full simulated result set is tabulated below.

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P001	Moderate-Risk	0.351	Deteriorating	Monitor
P002	High-Risk	0.256	Deteriorating	Monitor
P003	Low-Risk	0.122	Stable	Discharge
P004	Moderate-Risk	0.193	Critical	Escalate
P005	High-Risk	0.601	Critical	Escalate
P006	Low-Risk	0.038	Stable	Discharge
P007	Moderate-Risk	0.336	Deteriorating	Monitor
P008	High-Risk	0.866	Critical	Escalate
P009	Low-Risk	0.000	Stable	Discharge
P010	Moderate-Risk	0.285	Critical	Escalate
P011	High-Risk	0.690	Stable	Escalate
P012	Low-Risk	0.000	Stable	Discharge
P013	Moderate-Risk	0.391	Deteriorating	Monitor
P014	High-Risk	0.738	Critical	Escalate
P015	Low-Risk	0.000	Stable	Discharge
P016	Moderate-Risk	0.376	Deteriorating	Monitor
P017	High-Risk	0.939	Deteriorating	Escalate
P018	Low-Risk	0.107	Stable	Discharge
P019	Moderate-Risk	0.457	Deteriorating	Monitor
P020	High-Risk	0.775	Critical	Escalate
P021	Low-Risk	0.084	Stable	Discharge
P022	Moderate-Risk	0.446	Stable	Monitor
P023	High-Risk	0.527	Deteriorating	Monitor

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P024	Low-Risk	0.090	Stable	Discharge
P025	Moderate-Risk	0.194	Deteriorating	Monitor
P026	High-Risk	0.923	Critical	Escalate
P027	Low-Risk	0.171	Stable	Discharge
P028	Moderate-Risk	0.397	Deteriorating	Monitor
P029	High-Risk	0.603	Critical	Escalate
P030	Low-Risk	0.141	Deteriorating	Monitor
P031	Moderate-Risk	0.356	Deteriorating	Monitor
P032	High-Risk	0.618	Critical	Escalate
P033	Low-Risk	0.045	Deteriorating	Monitor
P034	Moderate-Risk	0.285	Stable	Discharge
P035	High-Risk	0.467	Stable	Monitor
P036	Low-Risk	0.050	Stable	Discharge
P037	Moderate-Risk	0.285	Deteriorating	Monitor
P038	High-Risk	0.788	Critical	Escalate
P039	Low-Risk	0.097	Stable	Discharge
P040	Moderate-Risk	0.456	Stable	Monitor
P041	High-Risk	0.496	Deteriorating	Monitor
P042	Low-Risk	0.008	Stable	Discharge
P043	Moderate-Risk	0.467	Stable	Monitor
P044	High-Risk	0.527	Deteriorating	Monitor
P045	Low-Risk	0.011	Stable	Discharge
P046	Moderate-Risk	0.241	Deteriorating	Monitor
P047	High-Risk	0.498	Stable	Monitor
P048	Low-Risk	0.104	Stable	Discharge
P049	Moderate-Risk	0.452	Critical	Escalate
P050	High-Risk	0.712	Stable	Escalate
P051	Low-Risk	0.132	Stable	Discharge
P052	Moderate-Risk	0.186	Stable	Discharge
P053	High-Risk	1.000	Critical	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P054	Low-Risk	0.029	Stable	Discharge
P055	Moderate-Risk	0.109	Stable	Discharge
P056	High-Risk	0.529	Critical	Escalate
P057	Low-Risk	0.021	Stable	Discharge
P058	Moderate-Risk	0.163	Deteriorating	Monitor
P059	High-Risk	0.238	Critical	Escalate
P060	Low-Risk	0.144	Stable	Discharge
P061	Moderate-Risk	0.173	Stable	Discharge
P062	High-Risk	0.917	Deteriorating	Escalate
P063	Low-Risk	0.078	Stable	Discharge
P064	Moderate-Risk	0.118	Stable	Discharge
P065	High-Risk	0.587	Deteriorating	Monitor
P066	Low-Risk	0.053	Stable	Discharge
P067	Moderate-Risk	0.397	Stable	Monitor
P068	High-Risk	0.745	Deteriorating	Escalate
P069	Low-Risk	0.044	Stable	Discharge
P070	Moderate-Risk	0.361	Stable	Monitor
P071	High-Risk	0.472	Critical	Escalate
P072	Low-Risk	0.026	Stable	Discharge
P073	Moderate-Risk	0.320	Stable	Monitor
P074	High-Risk	0.513	Deteriorating	Monitor
P075	Low-Risk	0.000	Stable	Discharge
P076	Moderate-Risk	0.395	Critical	Escalate
P077	High-Risk	0.553	Deteriorating	Monitor
P078	Low-Risk	0.048	Stable	Discharge
P079	Moderate-Risk	0.630	Critical	Escalate
P080	High-Risk	0.949	Critical	Escalate
P081	Low-Risk	0.145	Stable	Discharge
P082	Moderate-Risk	0.523	Stable	Monitor
P083	High-Risk	0.568	Critical	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P084	Low-Risk	0.073	Stable	Discharge
P085	Moderate-Risk	0.318	Deteriorating	Monitor
P086	High-Risk	0.404	Critical	Escalate
P087	Low-Risk	0.116	Stable	Discharge
P088	Moderate-Risk	0.276	Deteriorating	Monitor
P089	High-Risk	0.861	Stable	Escalate
P090	Low-Risk	0.080	Stable	Discharge
P091	Moderate-Risk	0.281	Stable	Discharge
P092	High-Risk	0.789	Critical	Escalate
P093	Low-Risk	0.086	Deteriorating	Monitor
P094	Moderate-Risk	0.364	Deteriorating	Monitor
P095	High-Risk	0.687	Critical	Escalate
P096	Low-Risk	0.122	Stable	Discharge
P097	Moderate-Risk	0.216	Deteriorating	Monitor
P098	High-Risk	0.727	Deteriorating	Escalate
P099	Low-Risk	0.217	Stable	Discharge
P100	Moderate-Risk	0.474	Critical	Escalate
P101	High-Risk	0.445	Deteriorating	Monitor
P102	Low-Risk	0.096	Stable	Discharge
P103	Moderate-Risk	0.573	Deteriorating	Monitor
P104	High-Risk	0.419	Stable	Monitor
P105	Low-Risk	0.099	Stable	Discharge
P106	Moderate-Risk	0.417	Deteriorating	Monitor
P107	High-Risk	0.936	Critical	Escalate
P108	Low-Risk	0.000	Stable	Discharge
P109	Moderate-Risk	0.104	Deteriorating	Monitor
P110	High-Risk	0.398	Critical	Escalate
P111	Low-Risk	0.000	Stable	Discharge
P112	Moderate-Risk	0.357	Deteriorating	Monitor
P113	High-Risk	0.726	Deteriorating	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P114	Low-Risk	0.158	Stable	Discharge
P115	Moderate-Risk	0.279	Deteriorating	Monitor
P116	High-Risk	1.000	Critical	Escalate
P117	Low-Risk	0.059	Critical	Escalate
P118	Moderate-Risk	0.324	Critical	Escalate
P119	High-Risk	0.462	Critical	Escalate
P120	Low-Risk	0.130	Stable	Discharge
P121	Moderate-Risk	0.459	Deteriorating	Monitor
P122	High-Risk	0.570	Critical	Escalate
P123	Low-Risk	0.178	Stable	Discharge
P124	Moderate-Risk	0.367	Stable	Monitor
P125	High-Risk	0.732	Critical	Escalate
P126	Low-Risk	0.127	Stable	Discharge
P127	Moderate-Risk	0.442	Deteriorating	Monitor
P128	High-Risk	0.476	Deteriorating	Monitor
P129	Low-Risk	0.048	Stable	Discharge
P130	Moderate-Risk	0.411	Stable	Monitor
P131	High-Risk	0.645	Critical	Escalate
P132	Low-Risk	0.119	Stable	Discharge
P133	Moderate-Risk	0.281	Deteriorating	Monitor
P134	High-Risk	0.631	Critical	Escalate
P135	Low-Risk	0.154	Stable	Discharge
P136	Moderate-Risk	0.411	Stable	Monitor
P137	High-Risk	1.000	Critical	Escalate
P138	Low-Risk	0.143	Stable	Discharge
P139	Moderate-Risk	0.400	Stable	Monitor
P140	High-Risk	0.693	Deteriorating	Escalate
P141	Low-Risk	0.064	Stable	Discharge
P142	Moderate-Risk	0.322	Deteriorating	Monitor
P143	High-Risk	0.289	Deteriorating	Monitor

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P144	Low-Risk	0.107	Stable	Discharge
P145	Moderate-Risk	0.514	Deteriorating	Monitor
P146	High-Risk	0.644	Critical	Escalate
P147	Low-Risk	0.151	Deteriorating	Monitor
P148	Moderate-Risk	0.250	Critical	Escalate
P149	High-Risk	1.000	Stable	Escalate
P150	Low-Risk	0.040	Stable	Discharge
P151	Moderate-Risk	0.540	Deteriorating	Monitor
P152	High-Risk	0.750	Critical	Escalate
P153	Low-Risk	0.000	Stable	Discharge
P154	Moderate-Risk	0.161	Stable	Discharge
P155	High-Risk	0.838	Stable	Escalate
P156	Low-Risk	0.169	Stable	Discharge
P157	Moderate-Risk	0.068	Deteriorating	Monitor
P158	High-Risk	0.711	Critical	Escalate
P159	Low-Risk	0.155	Critical	Escalate
P160	Moderate-Risk	0.366	Deteriorating	Monitor
P161	High-Risk	0.658	Deteriorating	Escalate
P162	Low-Risk	0.118	Stable	Discharge
P163	Moderate-Risk	0.597	Deteriorating	Monitor
P164	High-Risk	0.731	Stable	Escalate
P165	Low-Risk	0.039	Stable	Discharge
P166	Moderate-Risk	0.410	Stable	Monitor
P167	High-Risk	0.600	Critical	Escalate
P168	Low-Risk	0.000	Stable	Discharge
P169	Moderate-Risk	0.314	Stable	Monitor
P170	High-Risk	0.887	Deteriorating	Escalate
P171	Low-Risk	0.077	Stable	Discharge
P172	Moderate-Risk	0.520	Deteriorating	Monitor
P173	High-Risk	0.647	Critical	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P174	Low-Risk	0.206	Stable	Discharge
P175	Moderate-Risk	0.417	Stable	Monitor
P176	High-Risk	0.839	Deteriorating	Escalate
P177	Low-Risk	0.000	Stable	Discharge
P178	Moderate-Risk	0.181	Deteriorating	Monitor
P179	High-Risk	0.500	Deteriorating	Monitor
P180	Low-Risk	0.114	Stable	Discharge
P181	Moderate-Risk	0.295	Deteriorating	Monitor
P182	High-Risk	0.678	Critical	Escalate
P183	Low-Risk	0.032	Stable	Discharge
P184	Moderate-Risk	0.580	Stable	Monitor
P185	High-Risk	1.000	Deteriorating	Escalate
P186	Low-Risk	0.100	Stable	Discharge
P187	Moderate-Risk	0.121	Stable	Discharge
P188	High-Risk	0.730	Critical	Escalate
P189	Low-Risk	0.064	Stable	Discharge
P190	Moderate-Risk	0.359	Stable	Monitor
P191	High-Risk	0.559	Critical	Escalate
P192	Low-Risk	0.047	Stable	Discharge
P193	Moderate-Risk	0.418	Deteriorating	Monitor
P194	High-Risk	0.447	Deteriorating	Monitor
P195	Low-Risk	0.000	Stable	Discharge
P196	Moderate-Risk	0.398	Stable	Monitor
P197	High-Risk	0.792	Critical	Escalate
P198	Low-Risk	0.097	Stable	Discharge
P199	Moderate-Risk	0.426	Deteriorating	Monitor
P200	High-Risk	0.602	Deteriorating	Escalate
P201	Low-Risk	0.117	Stable	Discharge
P202	Moderate-Risk	0.217	Deteriorating	Monitor
P203	High-Risk	0.644	Critical	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P204	Low-Risk	0.145	Stable	Discharge
P205	Moderate-Risk	0.132	Stable	Discharge
P206	High-Risk	0.760	Stable	Escalate
P207	Low-Risk	0.090	Stable	Discharge
P208	Moderate-Risk	0.199	Stable	Discharge
P209	High-Risk	0.713	Critical	Escalate
P210	Low-Risk	0.044	Deteriorating	Monitor
P211	Moderate-Risk	0.337	Stable	Monitor
P212	High-Risk	0.615	Deteriorating	Escalate
P213	Low-Risk	0.151	Stable	Discharge
P214	Moderate-Risk	0.361	Deteriorating	Monitor
P215	High-Risk	0.670	Deteriorating	Escalate
P216	Low-Risk	0.065	Deteriorating	Monitor
P217	Moderate-Risk	0.317	Stable	Monitor
P218	High-Risk	0.397	Deteriorating	Monitor
P219	Low-Risk	0.015	Stable	Discharge
P220	Moderate-Risk	0.347	Deteriorating	Monitor
P221	High-Risk	0.493	Critical	Escalate
P222	Low-Risk	0.105	Stable	Discharge
P223	Moderate-Risk	0.332	Deteriorating	Monitor
P224	High-Risk	0.823	Critical	Escalate
P225	Low-Risk	0.126	Stable	Discharge
P226	Moderate-Risk	0.225	Stable	Discharge
P227	High-Risk	0.567	Deteriorating	Monitor
P228	Low-Risk	0.089	Stable	Discharge
P229	Moderate-Risk	0.273	Stable	Discharge
P230	High-Risk	0.628	Deteriorating	Escalate
P231	Low-Risk	0.039	Stable	Discharge
P232	Moderate-Risk	0.350	Deteriorating	Monitor
P233	High-Risk	0.757	Critical	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P234	Low-Risk	0.079	Stable	Discharge
P235	Moderate-Risk	0.158	Stable	Discharge
P236	High-Risk	0.585	Deteriorating	Monitor
P237	Low-Risk	0.043	Stable	Discharge
P238	Moderate-Risk	0.406	Stable	Monitor
P239	High-Risk	0.704	Critical	Escalate
P240	Low-Risk	0.056	Stable	Discharge
P241	Moderate-Risk	0.429	Deteriorating	Monitor
P242	High-Risk	0.666	Stable	Escalate
P243	Low-Risk	0.129	Stable	Discharge
P244	Moderate-Risk	0.467	Stable	Monitor
P245	High-Risk	0.596	Critical	Escalate
P246	Low-Risk	0.106	Stable	Discharge
P247	Moderate-Risk	0.383	Critical	Escalate
P248	High-Risk	0.423	Critical	Escalate
P249	Low-Risk	0.034	Deteriorating	Monitor
P250	Moderate-Risk	0.350	Deteriorating	Monitor
P251	High-Risk	0.812	Stable	Escalate
P252	Low-Risk	0.089	Stable	Discharge
P253	Moderate-Risk	0.355	Deteriorating	Monitor
P254	High-Risk	0.734	Deteriorating	Escalate
P255	Low-Risk	0.126	Stable	Discharge
P256	Moderate-Risk	0.443	Critical	Escalate
P257	High-Risk	0.461	Stable	Monitor
P258	Low-Risk	0.078	Stable	Discharge
P259	Moderate-Risk	0.322	Deteriorating	Monitor
P260	High-Risk	0.809	Stable	Escalate
P261	Low-Risk	0.141	Stable	Discharge
P262	Moderate-Risk	0.289	Stable	Discharge
P263	High-Risk	0.629	Deteriorating	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P264	Low-Risk	0.065	Stable	Discharge
P265	Moderate-Risk	0.406	Stable	Monitor
P266	High-Risk	0.645	Critical	Escalate
P267	Low-Risk	0.079	Stable	Discharge
P268	Moderate-Risk	0.312	Critical	Escalate
P269	High-Risk	0.552	Deteriorating	Monitor
P270	Low-Risk	0.123	Critical	Escalate
P271	Moderate-Risk	0.316	Deteriorating	Monitor
P272	High-Risk	0.317	Critical	Escalate
P273	Low-Risk	0.129	Stable	Discharge
P274	Moderate-Risk	0.390	Critical	Escalate
P275	High-Risk	0.364	Deteriorating	Monitor
P276	Low-Risk	0.149	Stable	Discharge
P277	Moderate-Risk	0.165	Deteriorating	Monitor
P278	High-Risk	0.443	Critical	Escalate
P279	Low-Risk	0.098	Deteriorating	Monitor
P280	Moderate-Risk	0.356	Stable	Monitor
P281	High-Risk	0.484	Critical	Escalate
P282	Low-Risk	0.110	Stable	Discharge
P283	Moderate-Risk	0.504	Stable	Monitor
P284	High-Risk	0.760	Deteriorating	Escalate
P285	Low-Risk	0.117	Stable	Discharge
P286	Moderate-Risk	0.164	Deteriorating	Monitor
P287	High-Risk	0.750	Critical	Escalate
P288	Low-Risk	0.067	Stable	Discharge
P289	Moderate-Risk	0.418	Critical	Escalate
P290	High-Risk	0.813	Deteriorating	Escalate
P291	Low-Risk	0.047	Stable	Discharge
P292	Moderate-Risk	0.396	Deteriorating	Monitor
P293	High-Risk	0.663	Critical	Escalate

PatientID	Cohort	Simulated P(Sepsis)	Simulated HMM State	Decision
P294	Low-Risk	0.087	Stable	Discharge
P295	Moderate-Risk	0.297	Stable	Discharge
P296	High-Risk	0.843	Critical	Escalate
P297	Low-Risk	0.031	Stable	Discharge
P298	Moderate-Risk	0.336	Deteriorating	Monitor
P299	High-Risk	0.677	Critical	Escalate
P300	Low-Risk	0.135	Stable	Discharge

K.1 Aggregate Decision Distribution by Cohort

Cohort	Escalate	Monitor	Discharge	Escalation Rate
Low-Risk	3	8	89	3.0%
Moderate-Risk	13	72	15	13.0%
High-Risk	79	21	0	79.0%

Table K.1. Aggregate simulated decision distribution across the full 300-patient synthetic cohort.

As expected, the escalation rate rises monotonically from the Low-Risk cohort through to the High-Risk cohort, confirming at a full-population scale the same qualitative behaviour already validated at the individual-case level in Table E.5.1 and Appendix F: the structured decision rule consistently escalates a much larger proportion of High-Risk-cohort patients than Low-Risk-cohort patients, while still escalating a small minority of Moderate- and even Low-Risk patients whose simulated symptom/state evidence happens to be more severe than their overall vitals-based cohort would suggest — precisely the behaviour intended by combining an overall cohort/severity signal with patient-specific graphical-model evidence rather than relying on cohort membership alone to drive clinical decisions.